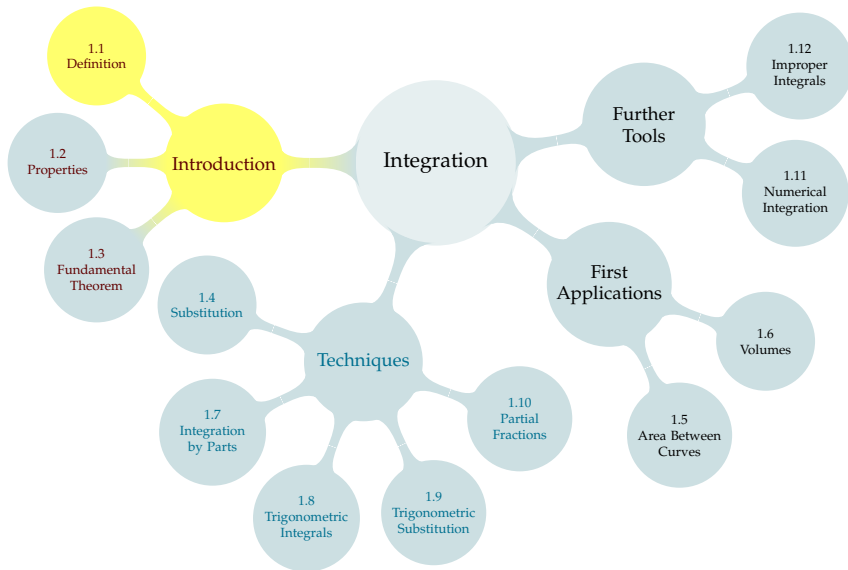


TABLE OF CONTENTS



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- ▶ Rate of change

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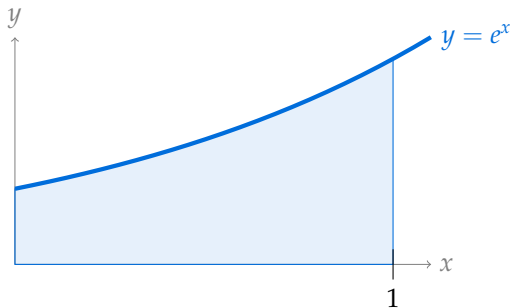
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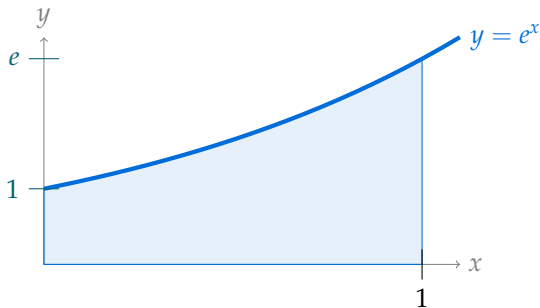
Integration

- ▶ Area under a curve
- ▶ “Reverse” of differentiation
- ▶ Solving differential equations
- ▶ Calculate net change from rate of change
- ▶ Volume of solids
- ▶ Work (in the physics sense)

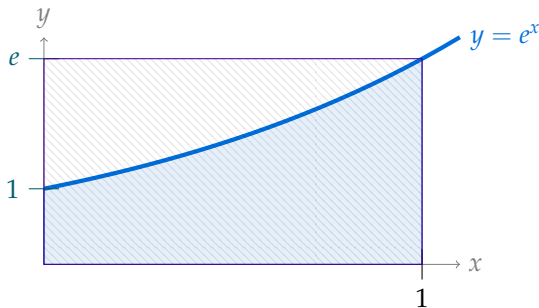
Approximate the area of the shaded region using rectangles.



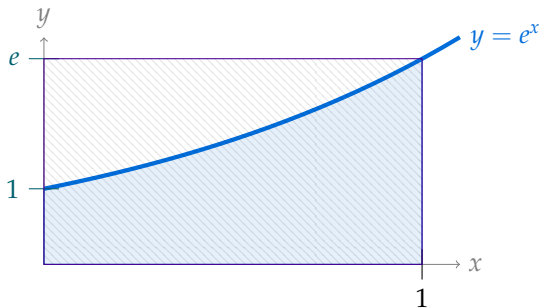
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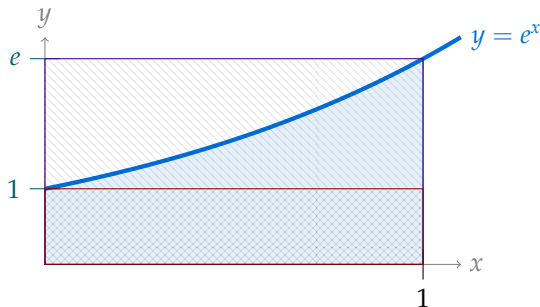


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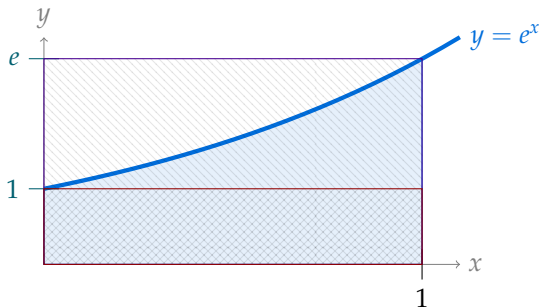
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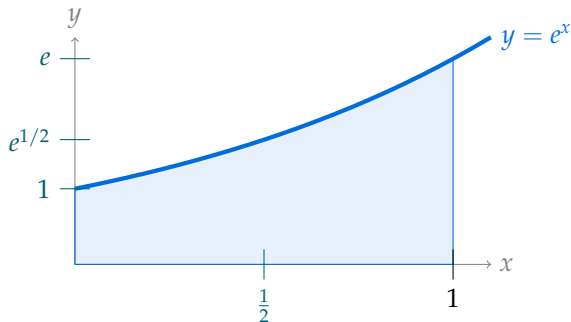
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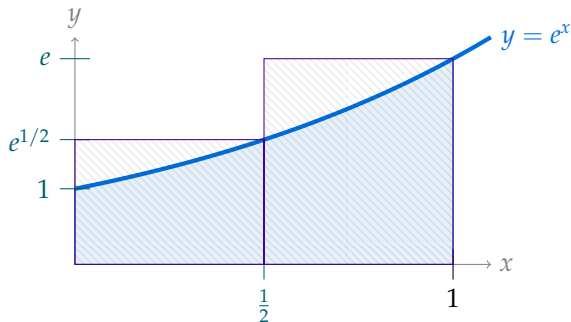
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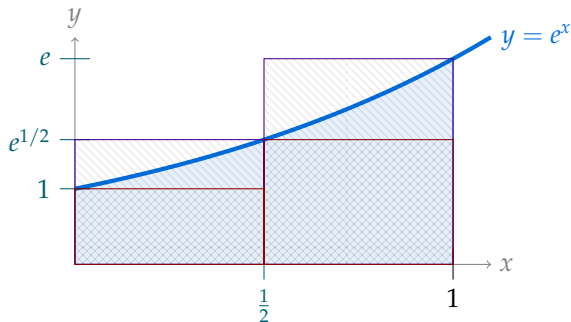
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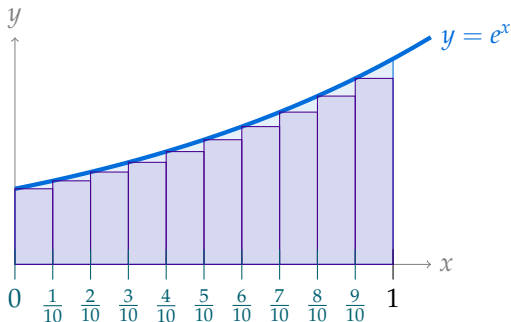
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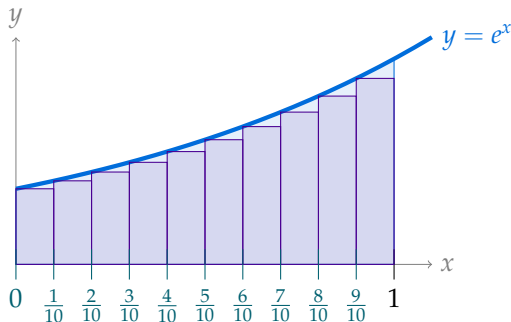


$$\left(\frac{1}{2} + \frac{1}{2}e^{1/2}\right) \leq \text{Area} \leq \left(\frac{1}{2}e^{1/2} + \frac{1}{2}e\right)$$

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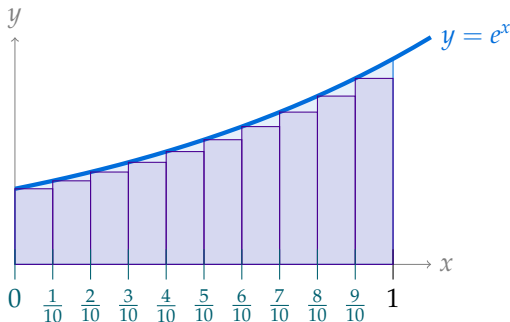


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$$\text{Area} \approx \frac{1}{10}(1) + \frac{1}{10} (e^{1/10}) + \frac{1}{10} (e^{2/10}) + \frac{1}{10} (e^{3/10}) + \cdots + \frac{1}{10} (e^{9/10})$$

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We're going to be doing a lot of adding.

SUMMATION (SIGMA) NOTATION

$$\sum_{i=a}^b f(i)$$

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$$\sum_{i=a}^b f(i) = f(a) + f(a+1) + f(a+2) + \cdots + f(b)$$

SIGMA NOTATION

Expand $\sum_{i=2}^4 (2i + 5).$

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$$\begin{aligned} \sum_{i=2}^4 (2i + 5) &= \underbrace{(2 \cdot 2 + 5)}_{i=2} + \underbrace{(2 \cdot 3 + 5)}_{i=3} + \underbrace{(2 \cdot 4 + 5)}_{i=4} \\ &= 9 + 11 + 13 = 33 \end{aligned}$$

SIGMA NOTATION

Expand $\sum_{i=1}^4 (i + (i - 1)^2).$

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$$\begin{aligned} &= \underbrace{(1 + 0^2)}_{i=1} + \underbrace{(2 + 1^2)}_{i=2} + \underbrace{(3 + 2^2)}_{i=3} + \underbrace{(4 + 3^2)}_{i=4} \\ &= 1 + 3 + 7 + 13 = 24 \end{aligned}$$

Write the following expressions in sigma notation:

► $3 + 4 + 5 + 6 + 7$

► $8 + 8 + 8 + 8 + 8$

► $1 + (-2) + 4 + (-8) + 16$

Write the following expressions in sigma notation:

► $3 + 4 + 5 + 6 + 7$
 $\sum_{i=3}^7 i$ and $\sum_{i=1}^5 (i+2)$ are two options (others are possible)

► $8 + 8 + 8 + 8 + 8$
 $\sum_{i=1}^5 8$ is one way (others are possible)

► $1 + (-2) + 4 + (-8) + 16$
 $\sum_{i=0}^4 (-2)^i$ is one way (others are possible)



ARITHMETIC OF SUMMATION NOTATION

Let c be a constant.

► Adding constants: $\sum_{i=1}^{10} c =$

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COMMON SUMS

Let $n \geq 1$ be an integer, a be a real number, and $r \neq 1$.

$$\sum_{i=0}^n ar^i = a + ar + ar^2 + \cdots + ar^n = a \frac{1 - r^{n+1}}{1 - r}$$

$$\sum_{i=1}^n i = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i^2 = 1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{i=1}^n i^3 = 1^3 + 2^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}$$

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$$\text{Simplify: } \sum_{i=1}^{13} (i^2 + i^3) = \sum_{i=1}^{13} i^2 + \sum_{i=1}^{13} i^3 = \frac{13(14)(27)}{6} + \frac{13^2(14^2)}{4}$$

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$$\sum_{i=1}^n i^3 = 1^3 + 2^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}$$

$$\text{Simplify: } \sum_{i=1}^{50} (1 - i^2) = \sum_{i=1}^{50} 1 - \sum_{i=1}^{50} i^2 = 50 - \frac{50(51)(101)}{6}$$

(OPTIONAL) PROOF OF A COMMON SUM

Here is a derivation of $\sum_{i=0}^n r^i = \frac{1 - r^{n+1}}{1 - r}$:

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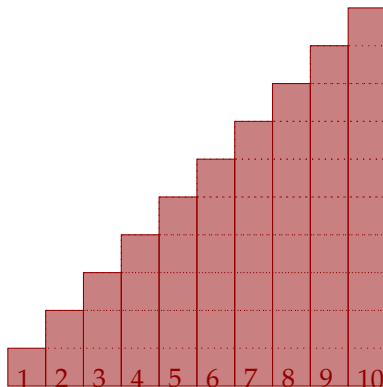
$$rA = \quad \cancel{r} + \cancel{r^2} + \cdots + \cancel{r^n} + \cancel{r^{n+1}} + r^{n+1}$$

subtract $A - rA = 1 - r^{n+1}$

divide across $A = \frac{1 - r^{n+1}}{1 - r}$

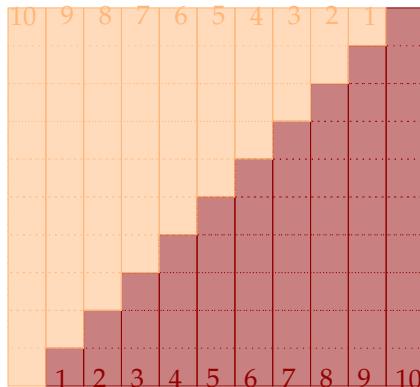
(OPTIONAL) PROOF OF ANOTHER COMMON SUM

$$\sum_{i=1}^{10} i = 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 =$$



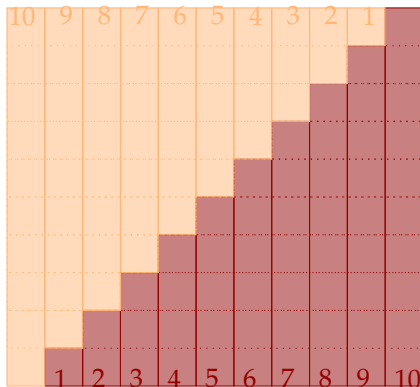
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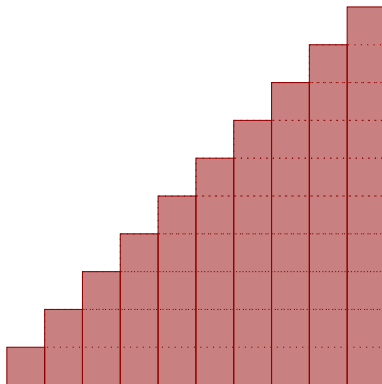
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$$\sum_{i=1}^{10} i = 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = \frac{10 \cdot 11}{2}$$



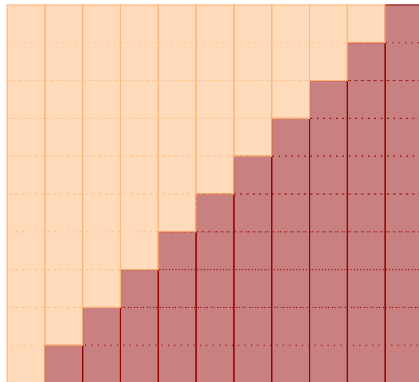
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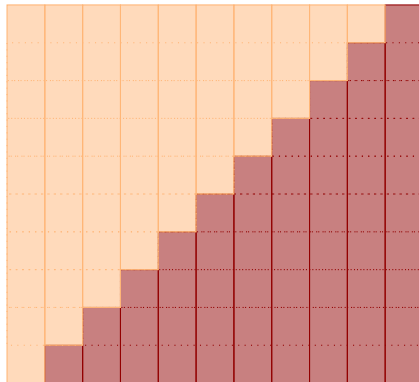
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Notation

The symbol

$$\int_a^b f(x) \, dx$$

is read “the definite integral of the function $f(x)$ from a to b .”

- ▶ $f(x)$: integrand
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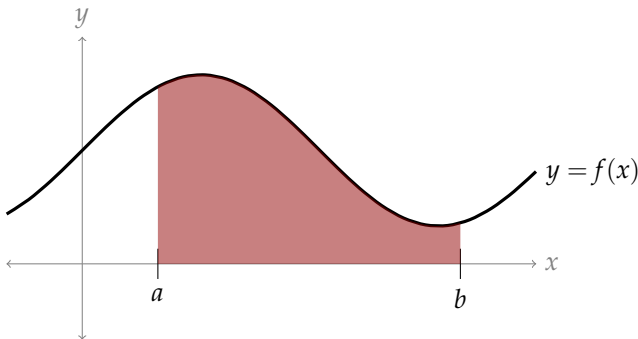
- ▶ $f(x)$: integrand
- ▶ a and b : limits of integration
- ▶ dx : differential

$$\int_a^b f(x) \, dx$$

A graph of a function $y = f(x)$ on a Cartesian coordinate system. The x-axis has two points marked a and b . The function curve starts at a low point, crosses the x-axis, reaches a local maximum, crosses the x-axis again, reaches a local minimum, and then crosses the x-axis a third time.

$$\int_a^b f(x) \, dx$$

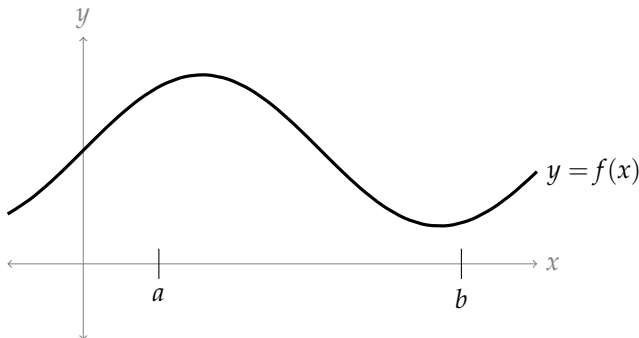
is “the area of the region bounded above by $y = f(x)$, below by $y = 0$, to the left by $x = a$, and to the right by $x = b$.”



If $f(x) \geq 0$ and $a \leq b$, one interpretation of

$$\int_a^b f(x) \, dx$$

is the **signed** area of the region between $y = f(x)$ and $y = 0$, from $x = a$ to $x = b$. Area **above** the axis is **positive**, and area **below** it is **negative**.



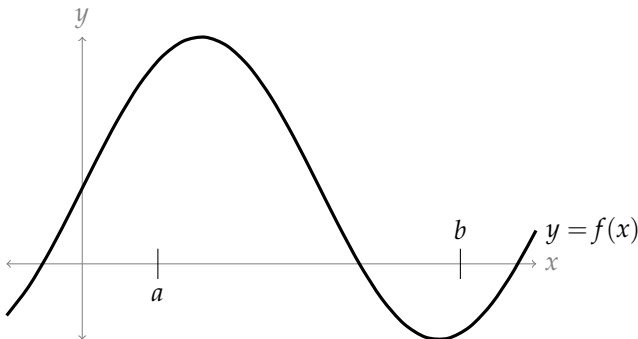
$$\int_a^b f(x) \, dx$$

A graph of a function $y = f(x)$ is shown on a Cartesian coordinate system. The x-axis is labeled x and the y-axis is labeled y . The curve $y = f(x)$ is a smooth, continuous function. The area under the curve between $x = a$ and $x = b$ is shaded in red, representing the definite integral of $f(x)$ from a to b .

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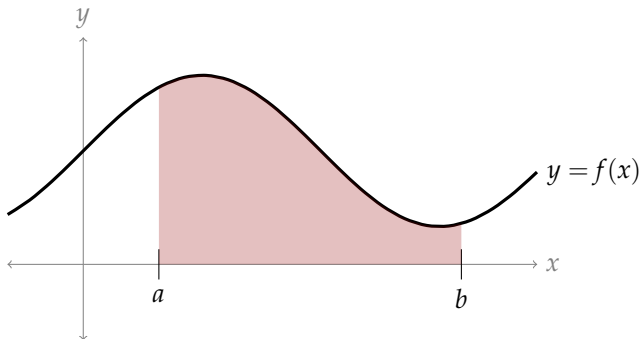


$$\int_a^b f(x) \, dx$$

A graph of a function $y = f(x)$ is shown on a Cartesian coordinate system. The x-axis is labeled x and the y-axis is labeled y . The curve $y = f(x)$ is plotted. The area under the curve is divided into two regions: a red region from $x = a$ to the first x-intercept, and a blue region from the first x-intercept to $x = b$. The red region is above the x-axis, and the blue region is below the x-axis.

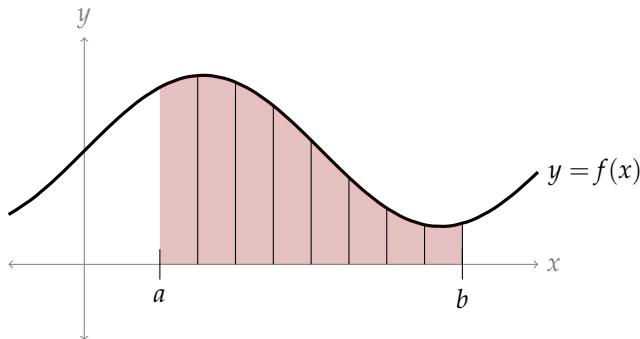
RIEMANN SUMS

A **Riemann sum** approximates the area under a curve by cutting it into equal-width segments, and approximating each segment as a rectangle.



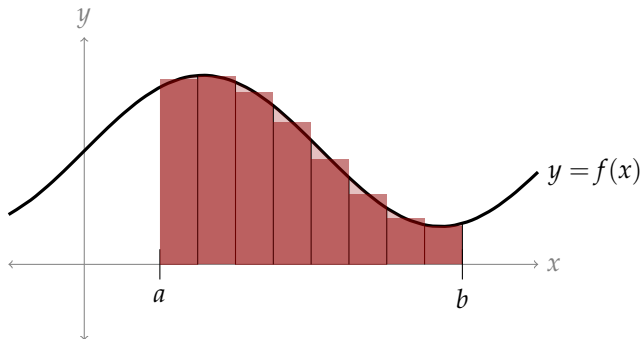
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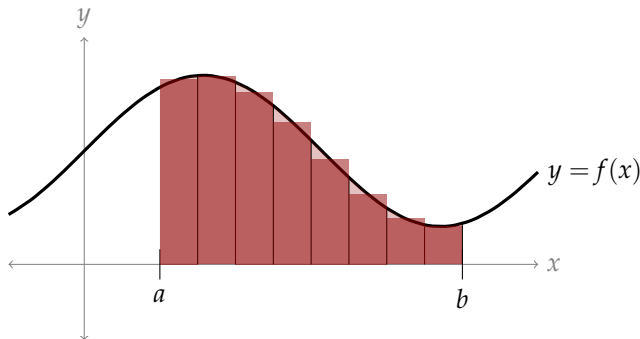
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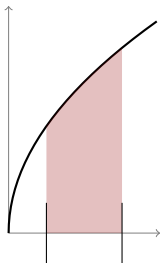
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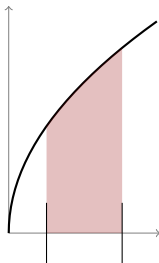


There are different ways to choose the height of each rectangle.

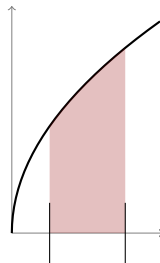
TYPES OF RIEMANN SUMS (RS)



Left RS

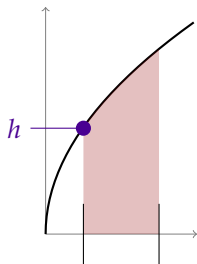


Right RS

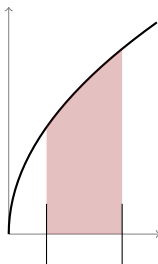


Midpoint RS

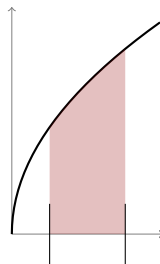
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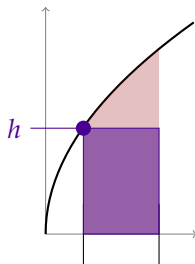


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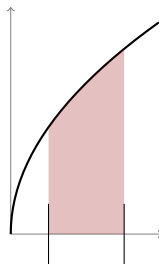


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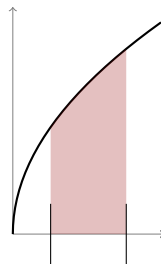
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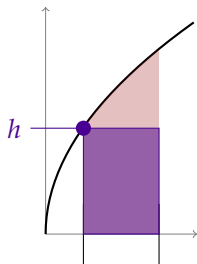


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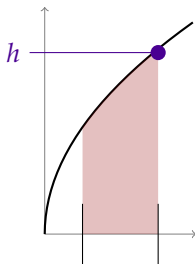


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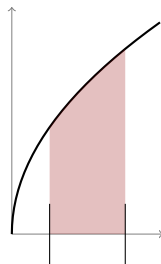
TYPES OF RIEMANN SUMS (RS)



Left RS

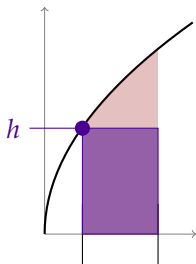


Right RS

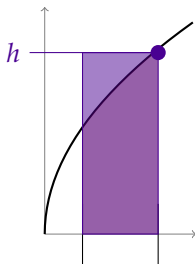


Midpoint RS

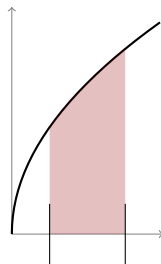
TYPES OF RIEMANN SUMS (RS)



Left RS

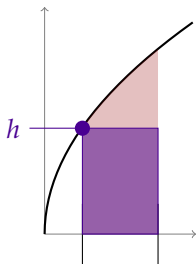


Right RS

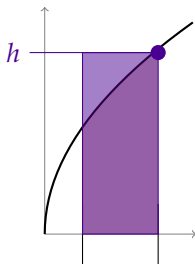


Midpoint RS

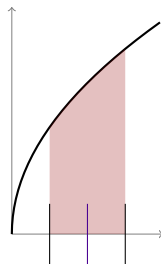
TYPES OF RIEMANN SUMS (RS)



Left RS

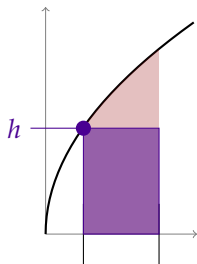


Right RS

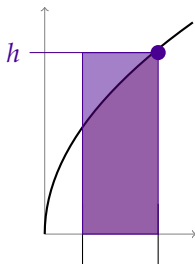


Midpoint RS

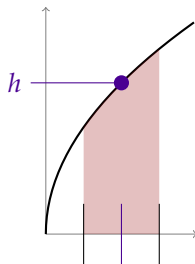
TYPES OF RIEMANN SUMS (RS)



Left RS

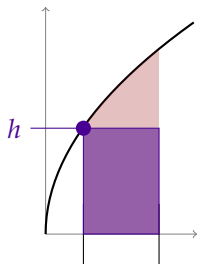


Right RS

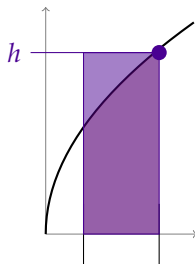


Midpoint RS

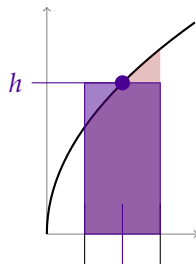
TYPES OF RIEMANN SUMS (RS)



Left RS

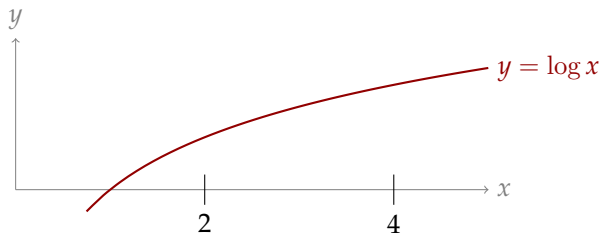


Right RS

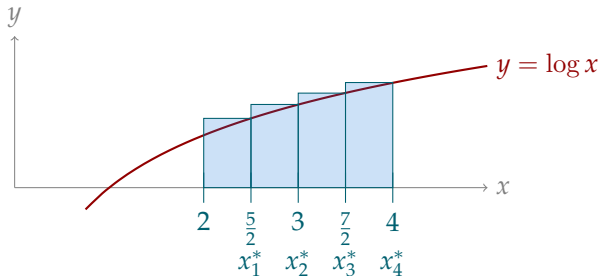


Midpoint RS

Approximate $\int_2^4 \log(x) \, dx$ using a **right Riemann sum** with $n = 4$ rectangles. For now, do not use sigma notation.



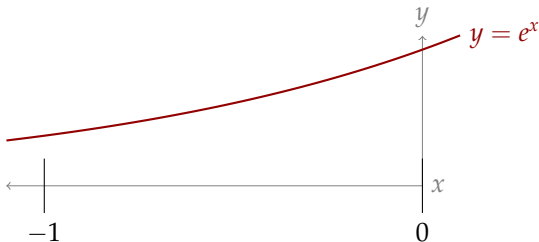
Approximate $\int_2^4 \log(x) dx$ using a **right Riemann sum** with $n = 4$ rectangles. For now, do not use sigma notation.



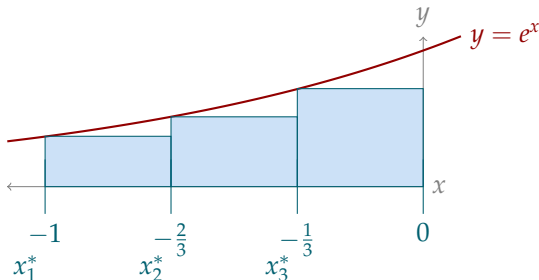
- Width of each rectangle: $\frac{4-2}{4} = \frac{1}{2}$
- Heights taken at right endpoints of rectangles:
 $x_1^* = \frac{5}{2}, x_2^* = 3, x_3^* = \frac{7}{2}, x_4^* = 4$

$$\int_2^4 \log(x) dx \approx \frac{1}{2} \log\left(\frac{5}{2}\right) + \frac{1}{2} \log(3) + \frac{1}{2} \log\left(\frac{7}{2}\right) + \frac{1}{2} \log(4)$$

Approximate $\int_{-1}^0 e^x dx$ using a **left Riemann sum** with $n = 3$ rectangles. For now, do not use sigma notation.



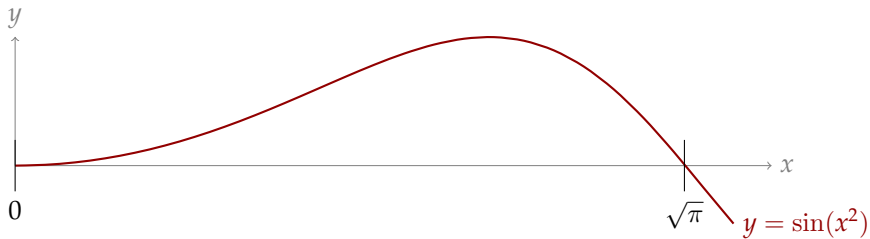
Approximate $\int_{-1}^0 e^x dx$ using a **left Riemann sum** with $n = 3$ rectangles. For now, do not use sigma notation.



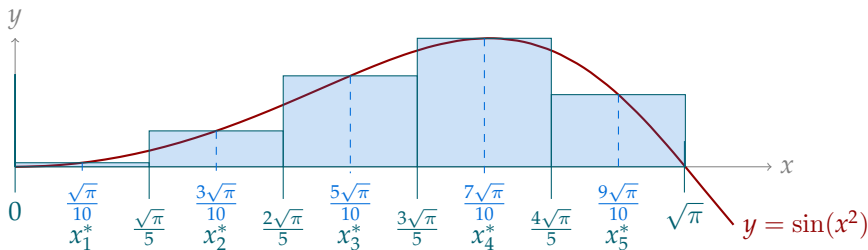
- Width of each rectangle: $\frac{0 - (-1)}{3} = \frac{1}{3}$
- Heights taken at left endpoints of rectangles:
 $x_1^* = -1, x_2^* = -\frac{2}{3}, x_3^* = -\frac{1}{3}$

$$\int_{-1}^0 e^x dx \approx \frac{1}{3}e^{-1} + \frac{1}{3}e^{-2/3} + \frac{1}{3}e^{-1/3}$$

Approximate $\int_0^{\sqrt{\pi}} \sin(x^2) dx$ using a **midpoint Riemann sum** with $n = 5$ rectangles. For now, do not use sigma notation.



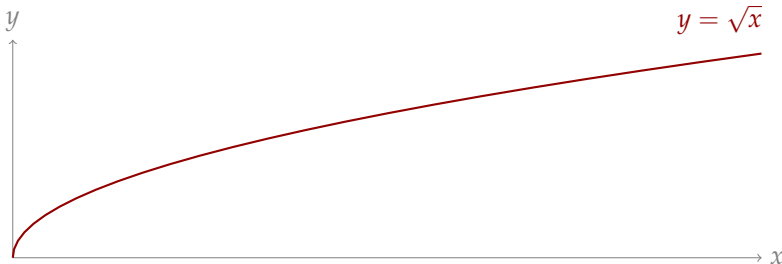
Approximate $\int_0^{\sqrt{\pi}} \sin(x^2) dx$ using a **midpoint Riemann sum** with $n = 5$ rectangles. For now, do not use sigma notation.



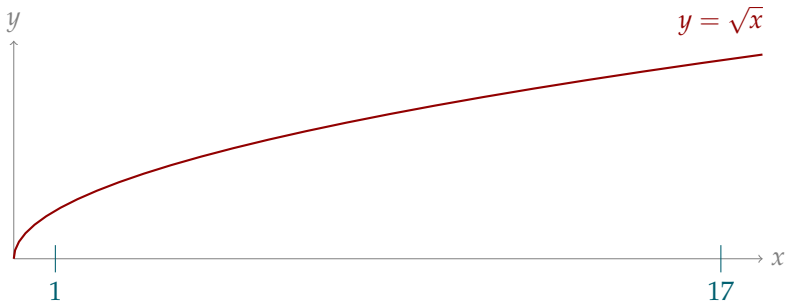
- Width of each rectangle: $\frac{\sqrt{\pi}-0}{5} = \frac{\sqrt{\pi}}{5}$
- Heights taken at midpoints of rectangles:
 $x_1^* = \frac{\sqrt{\pi}}{10}$, $x_2^* = \frac{3\sqrt{\pi}}{10}$, $x_3^* = \frac{5\sqrt{\pi}}{10}$, $x_4^* = \frac{7\sqrt{\pi}}{10}$, $x_5^* = \frac{9\sqrt{\pi}}{10}$

$$\frac{\sqrt{\pi}}{5} \left[\sin\left(\frac{\pi}{100}\right) + \sin\left(\frac{9\pi}{100}\right) + \sin\left(\frac{25\pi}{100}\right) + \sin\left(\frac{49\pi}{100}\right) + \sin\left(\frac{81\pi}{100}\right) \right]$$

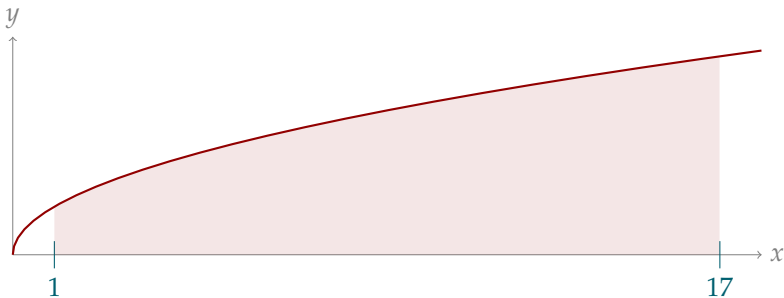
Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.



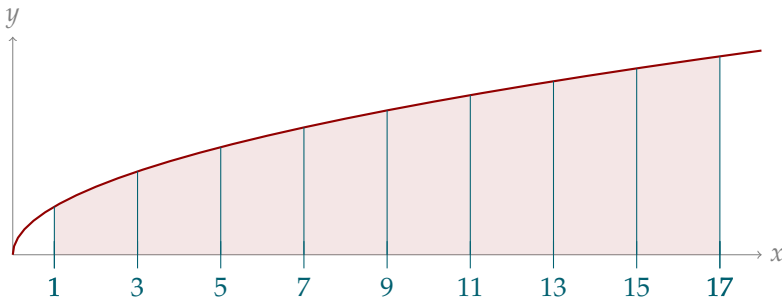
Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.



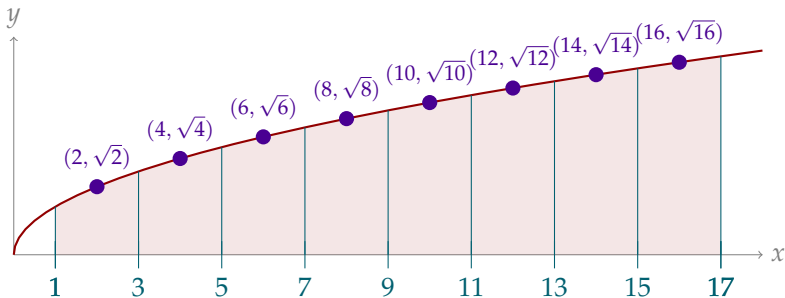
Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.



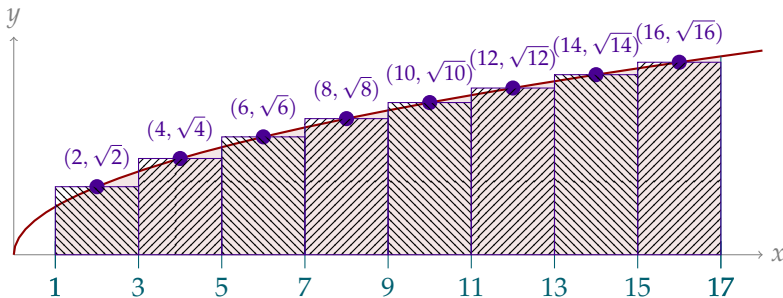
Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.



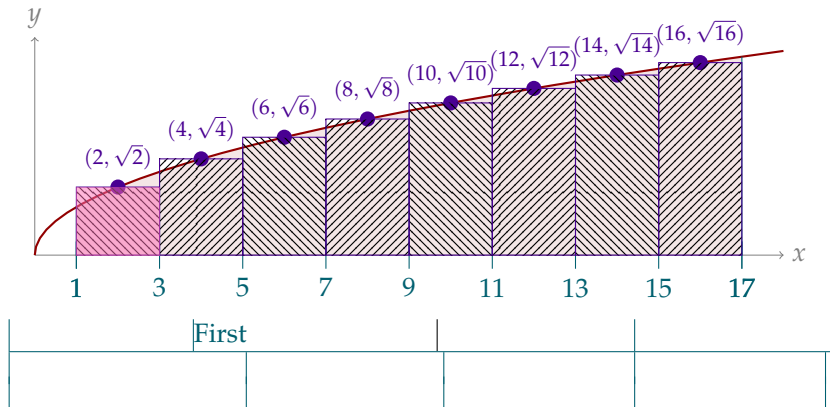
Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.



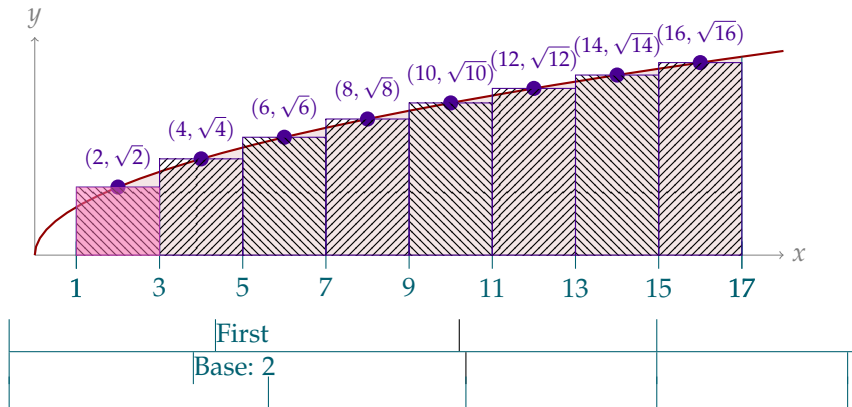
Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.



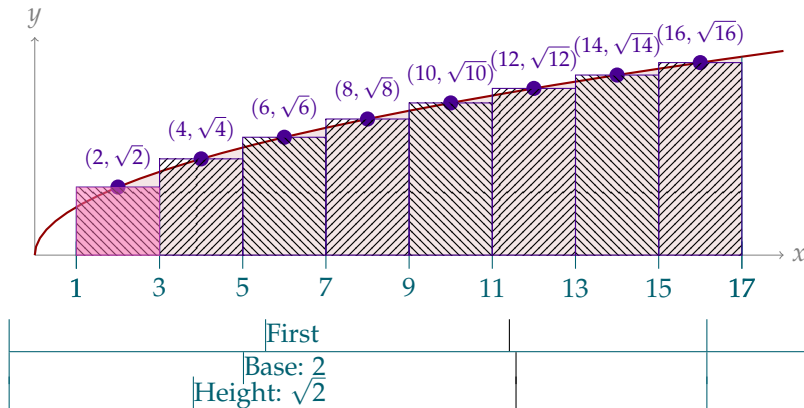
Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.



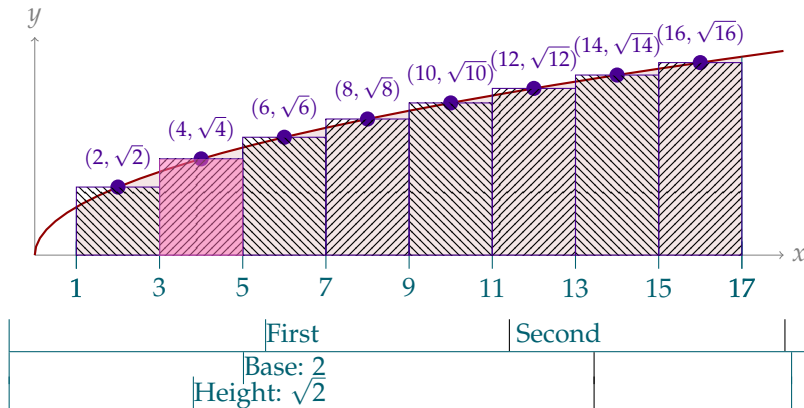
Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.



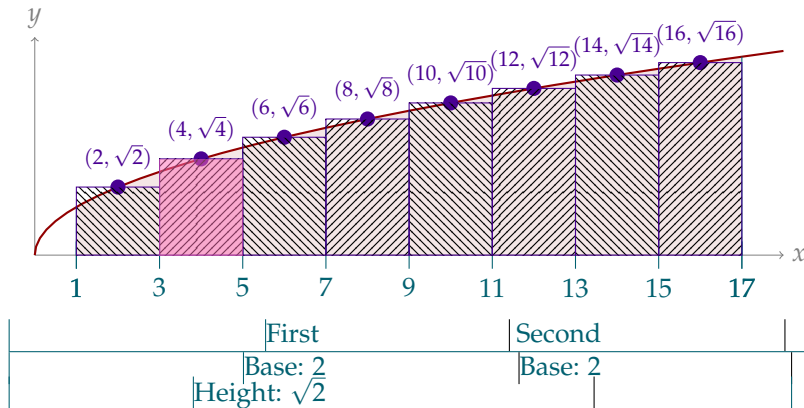
Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.



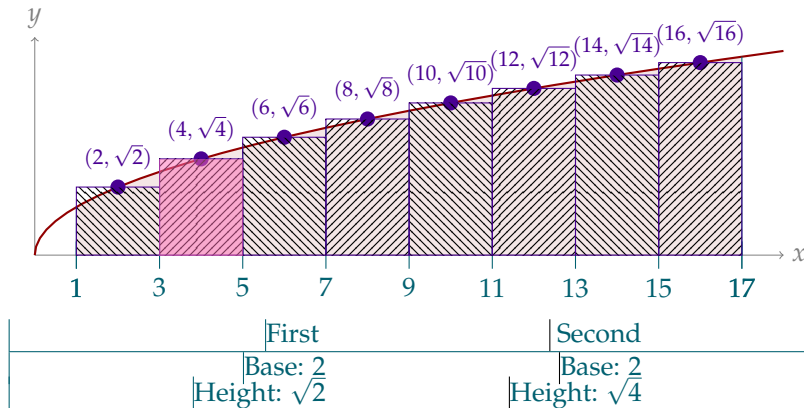
Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.



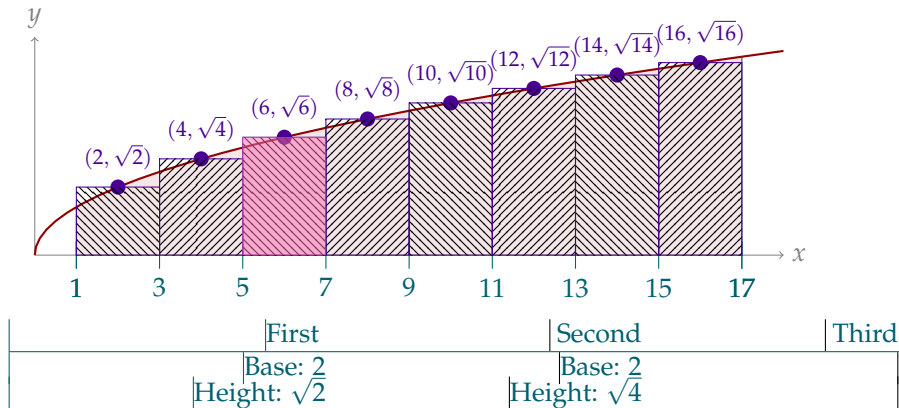
Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.



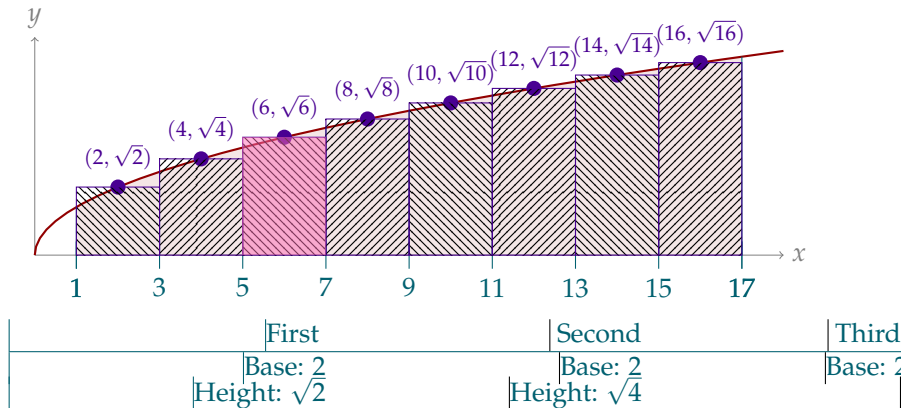
Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.



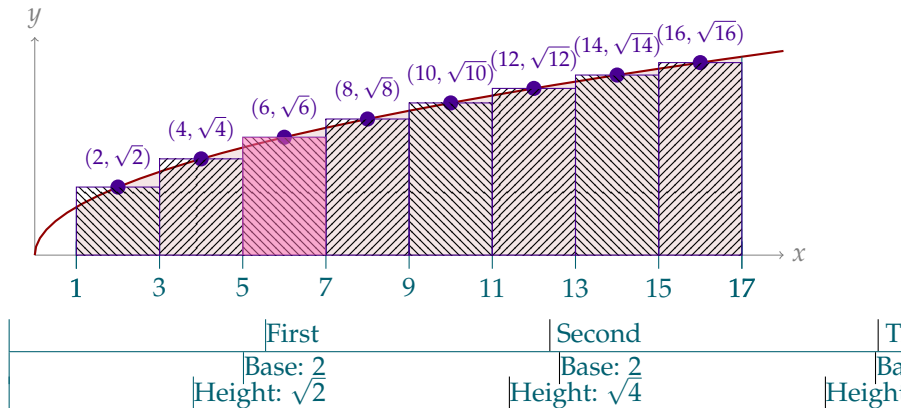
Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.



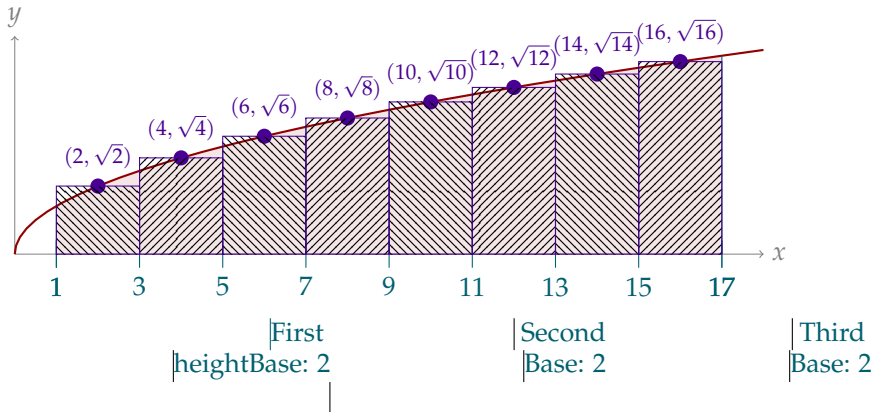
Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.



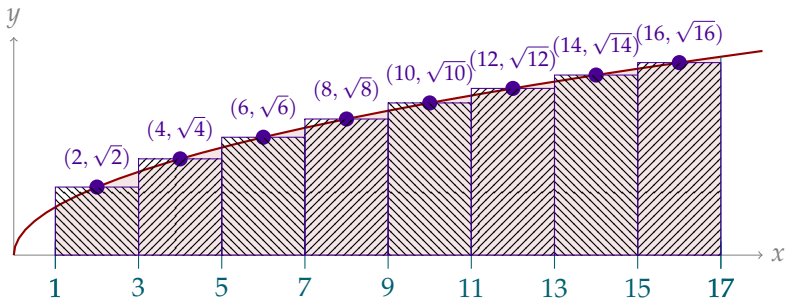
Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.



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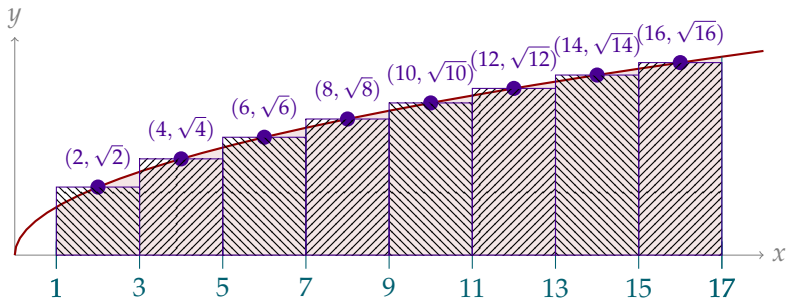


$i = 1$
heightBase: 2

Second
Base: 2

Third
Base: 2

Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.

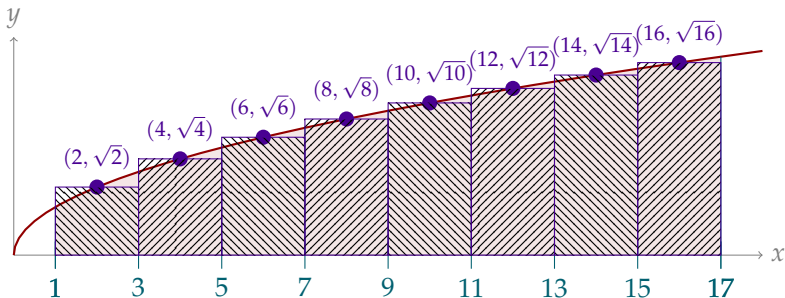


$i = 1$
heightBase: 2

$i = 2$
Base: 2

Third
Base: 2

Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.

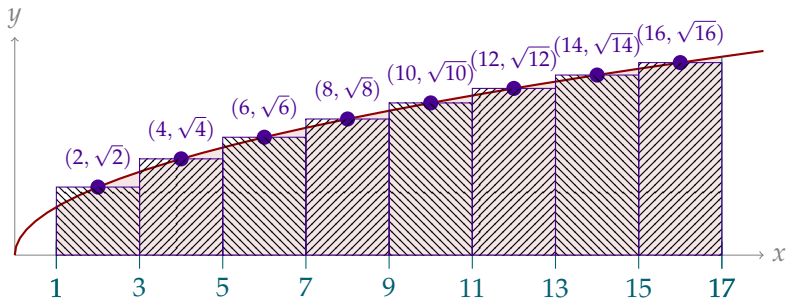


$i = 1$
heightBase: 2

$i = 2$
Base: 2

$i = 3$
Base: 2

Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.

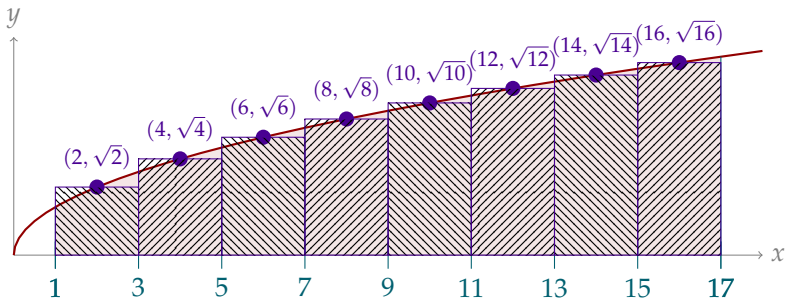


$i = 1$
heightBase: 2

$i = 2$
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$i = 3$
Base: 2

Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.

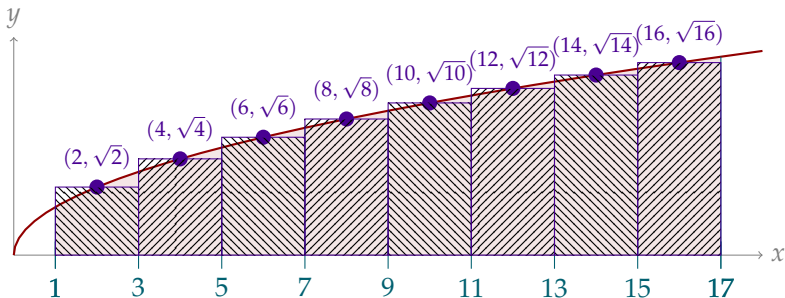


$i = 1$
heightBase: 2

$i = 2$
Base: 2

$i = 3$
Base: 2

Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.

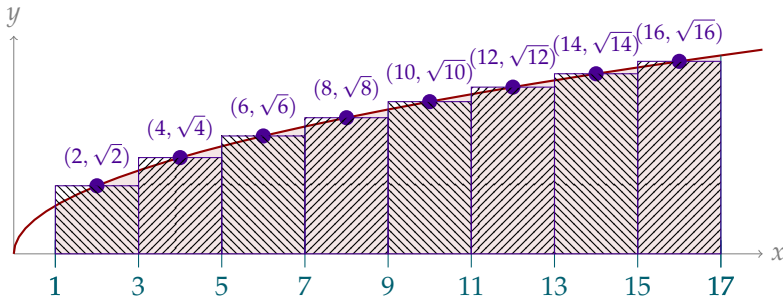


$i = 1$
heightBase: 2

$i = 2$
Base: 2

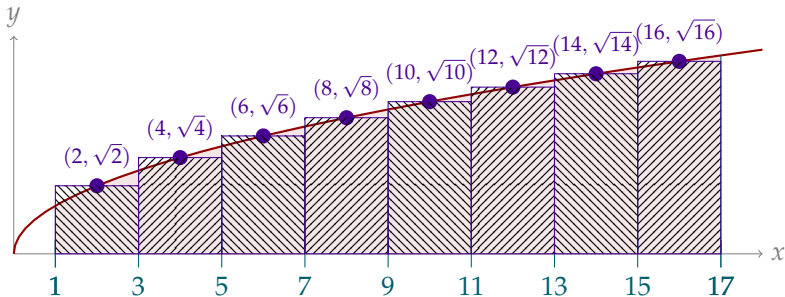
$i = 3$
Base: 2

Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.



The i th rectangle has base 2 and height $\sqrt{2i}$, so

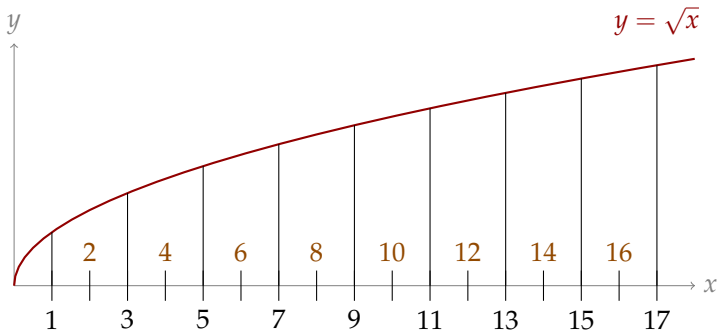
Approximate $\int_1^{17} \sqrt{x} \, dx$ using a **midpoint Riemann sum** with 8 rectangles. Write the result in sigma notation.



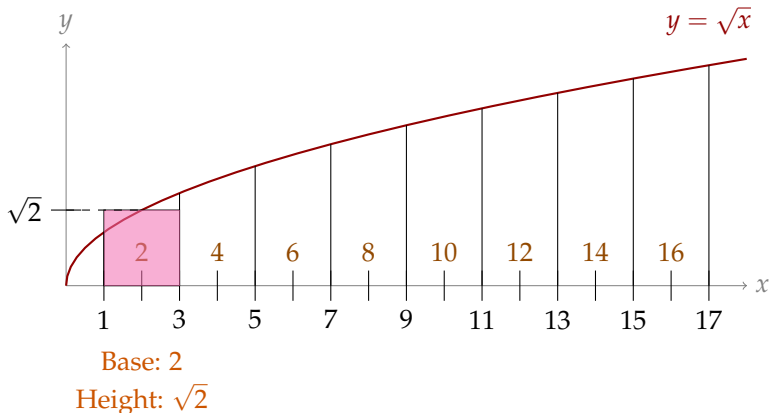
The i th rectangle has base 2 and height $\sqrt{2i}$, so

$$\text{area} \approx \sum_{i=1}^8 2\sqrt{2i}$$

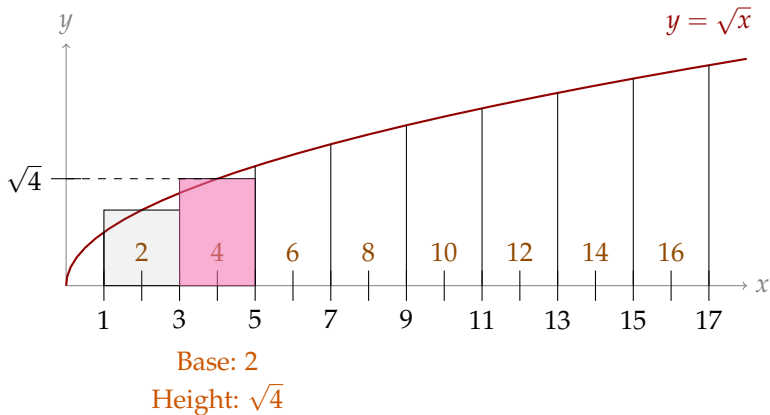
$$\sum_{i=1}^8 2\sqrt{2i} = \underbrace{2\sqrt{2}}_{i=1} + \underbrace{2\sqrt{4}}_{i=2} + \underbrace{2\sqrt{6}}_{i=3} + \underbrace{2\sqrt{8}}_{i=4} + \underbrace{2\sqrt{10}}_{i=5} + \underbrace{2\sqrt{12}}_{i=6} + \underbrace{2\sqrt{14}}_{i=7} + \underbrace{2\sqrt{16}}_{i=8}$$



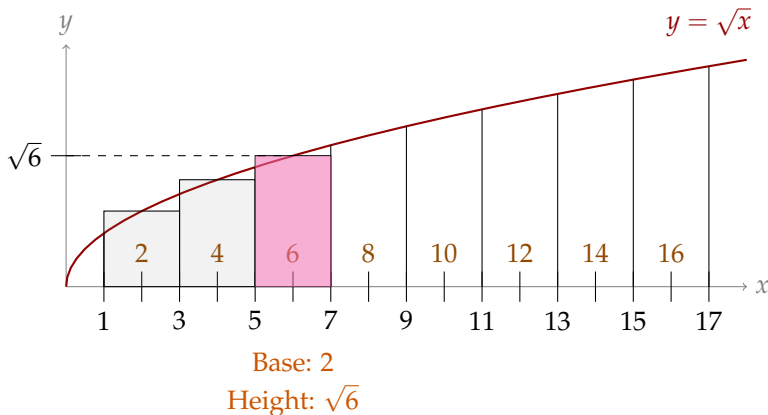
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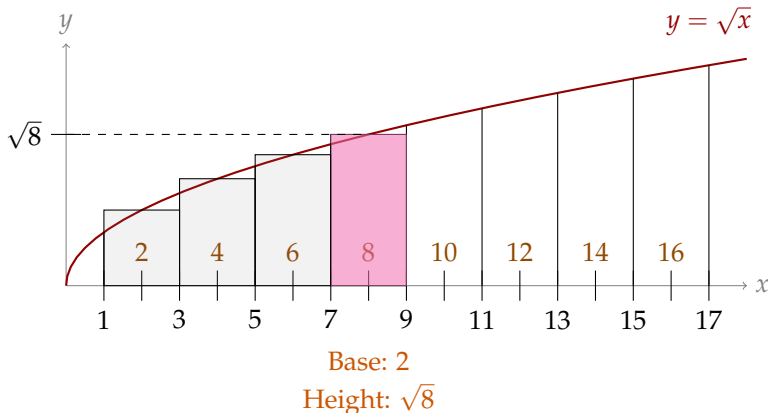
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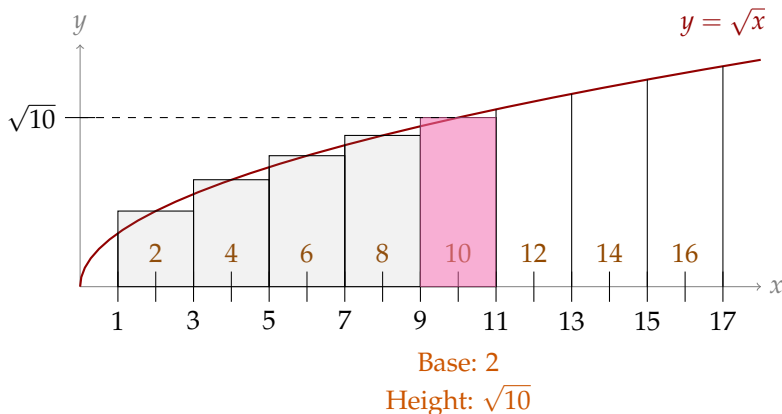
$$\sum_{i=1}^8 2\sqrt{2i} = \underbrace{2\sqrt{2}}_{i=1} + \underbrace{2\sqrt{4}}_{i=2} + \underbrace{2\sqrt{6}}_{i=3} + \underbrace{2\sqrt{8}}_{i=4} + \underbrace{2\sqrt{10}}_{i=5} + \underbrace{2\sqrt{12}}_{i=6} + \underbrace{2\sqrt{14}}_{i=7} + \underbrace{2\sqrt{16}}_{i=8}$$



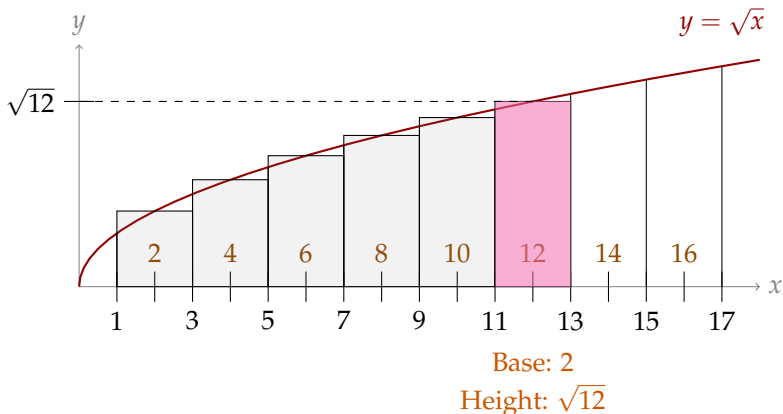
$$\sum_{i=1}^8 2\sqrt{2i} = \underbrace{2\sqrt{2}}_{i=1} + \underbrace{2\sqrt{4}}_{i=2} + \underbrace{2\sqrt{6}}_{i=3} + \underbrace{2\sqrt{8}}_{i=4} + \underbrace{2\sqrt{10}}_{i=5} + \underbrace{2\sqrt{12}}_{i=6} + \underbrace{2\sqrt{14}}_{i=7} + \underbrace{2\sqrt{16}}_{i=8}$$



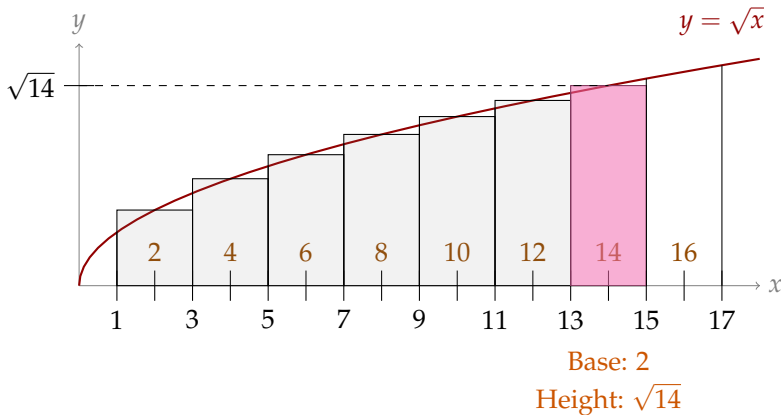
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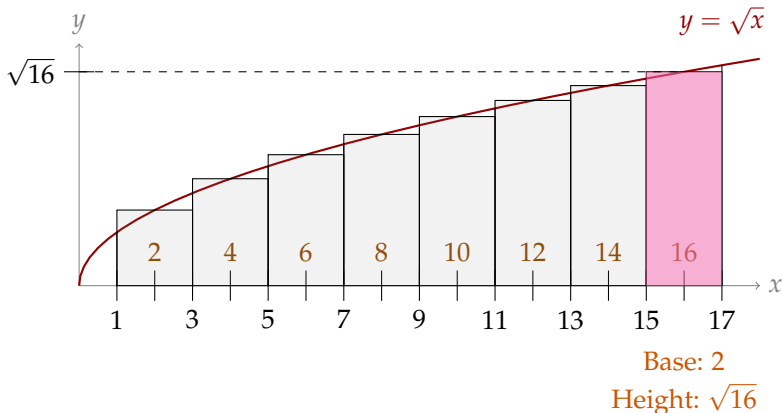
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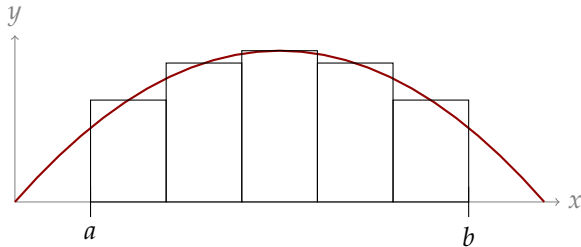


Riemann sum with n rectangles

$$\int_a^b f(x) \, dx \approx \sum_{i=1}^n \Delta x \cdot f(x_{i,n}^*)$$

where $\Delta x = \frac{b-a}{n}$ and $x_{i,n}^*$ is an x -value in the i th rectangle.

$$\sum_{i=1}^n \Delta x \cdot f(x_{i,n}^*) = \Delta x \cdot f(x_{1,n}^*) + \Delta x \cdot f(x_{2,n}^*) + \Delta x \cdot f(x_{3,n}^*) + \cdots + \Delta x \cdot f(x_{n,n}^*)$$

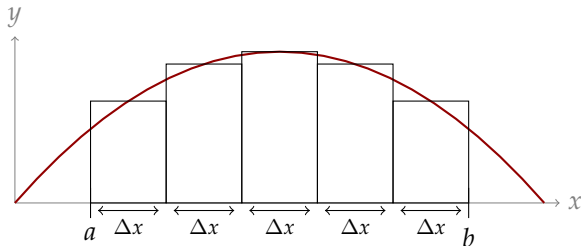


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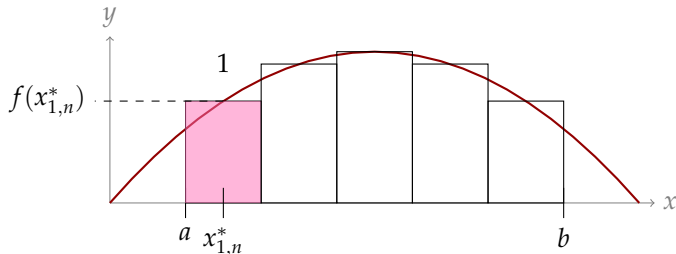


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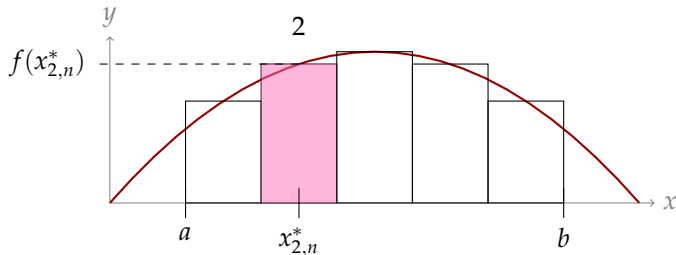


Riemann sum with n rectangles

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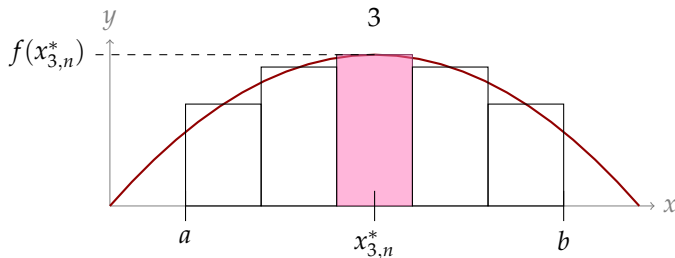


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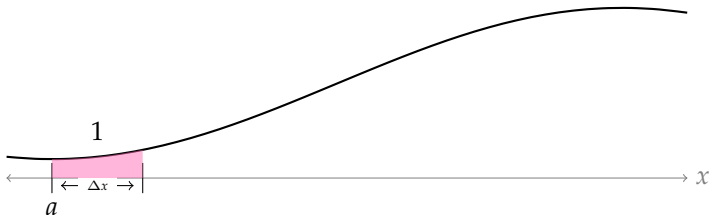
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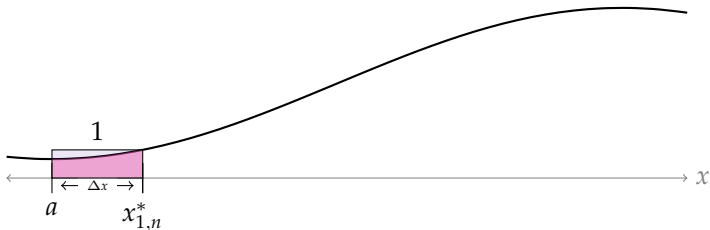
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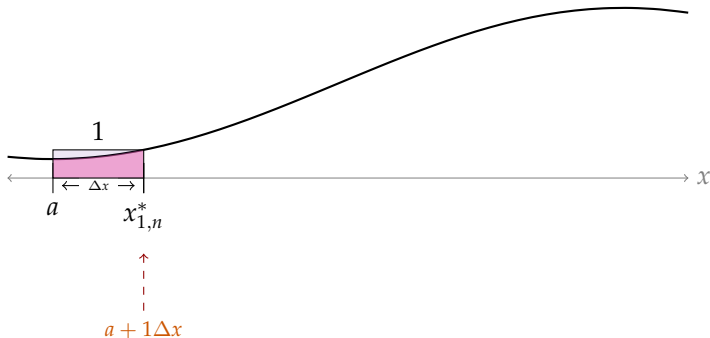
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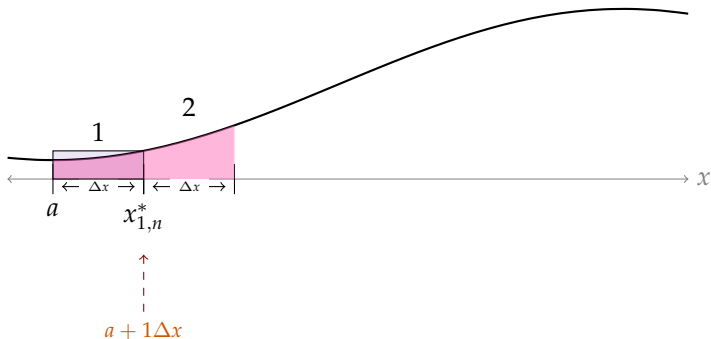
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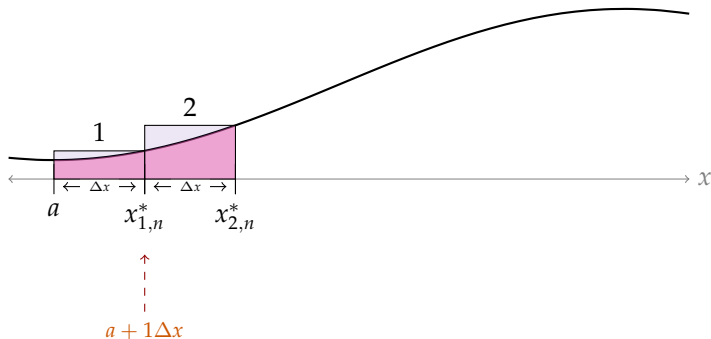
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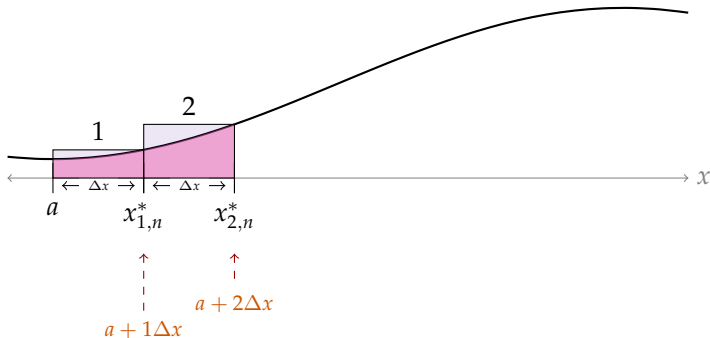
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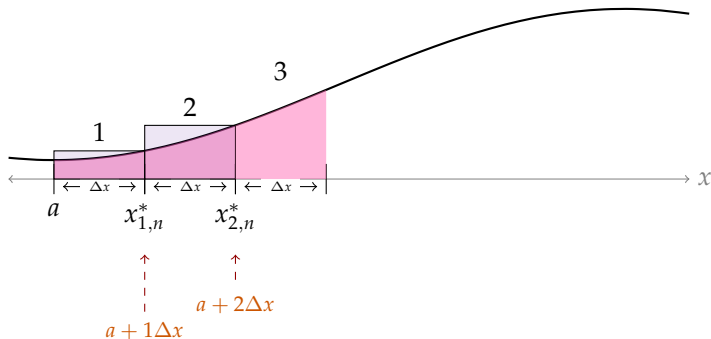
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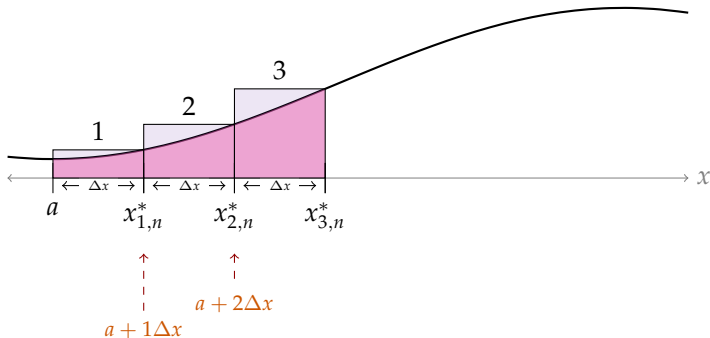
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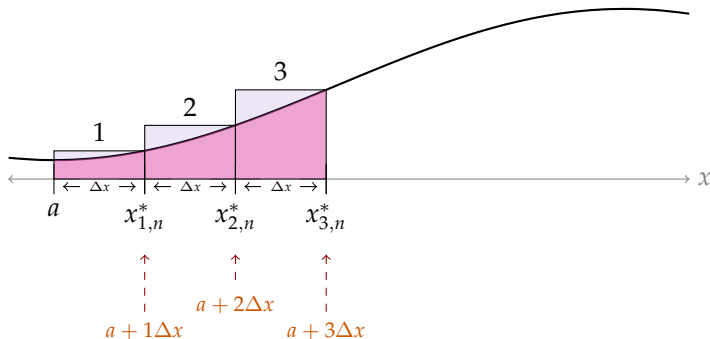
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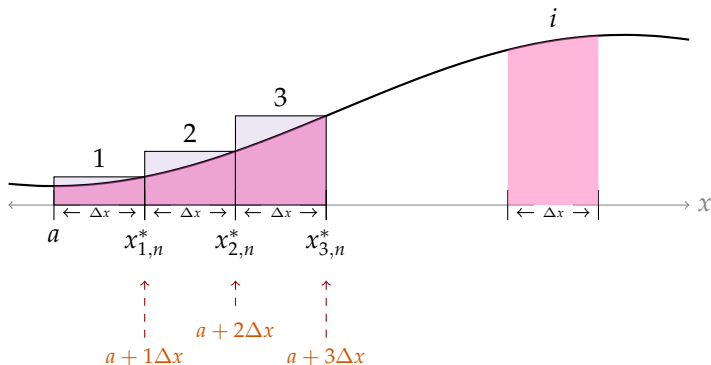
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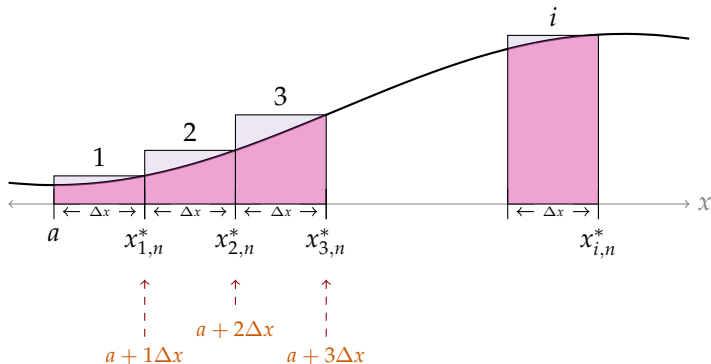
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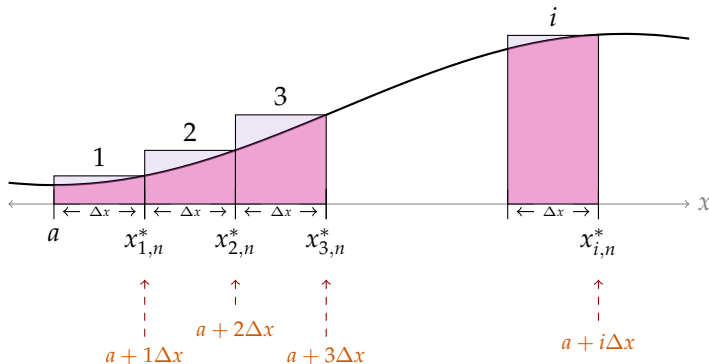
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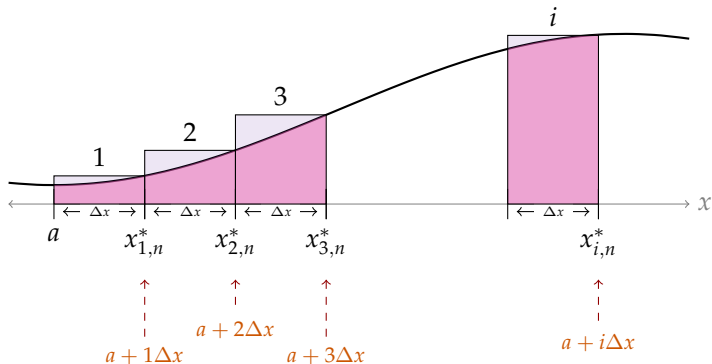
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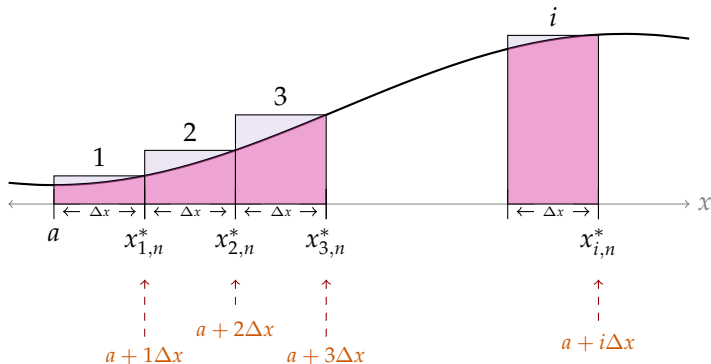
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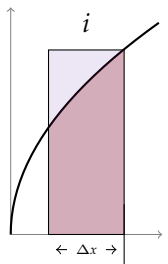
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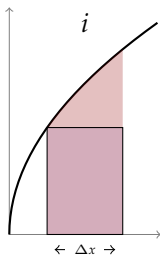
TYPES OF RIEMANN SUMS (RS)

What height would you choose for the i th rectangle?

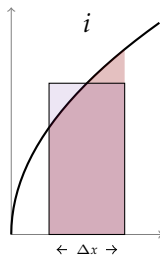


$$a + i\Delta x$$

Right RS



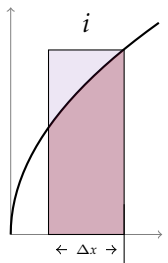
Left RS



Midpoint RS

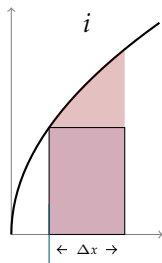
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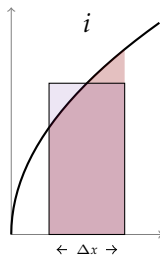
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$$a + (i - 1)\Delta x$$

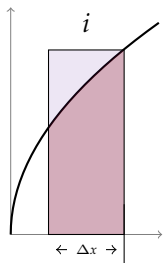
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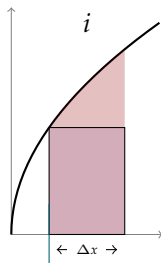
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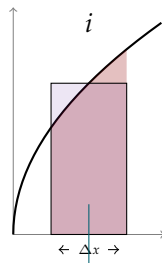
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Midpoint RS

Riemann sums with n rectangles. Let $\Delta x = \frac{b-a}{n}$

The **right** Riemann sum approximation of $\int_a^b f(x) \, dx$ is:

$$\sum_{i=1}^n \Delta x \cdot f(a + i\Delta x)$$

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Give a right Riemann Sum for the area under the curve $y = x^2 - x$ from $a = 1$ to $b = 6$ using $n = 1000$ intervals.

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$$\sum_{n=1}^{1000} \frac{5}{1000} \left[\left(1 + \frac{5}{1000}i \right)^2 - \left(1 + \frac{5}{1000}i \right) \right]$$

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$$\sum_{n=1}^{1000} \frac{3}{1000} \left[5 \left(6 + \frac{3}{1000} (i - 1/2) \right) - \left(6 + \frac{3}{1000} (i - 1/2) \right)^2 \right]$$

EVALUATING RIEMANN SUMS

[▶ SKIP RIEMANN EVALUATIONS](#)

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}$$

Give the right Riemann sum of $f(x) = x^2$ from $a = 0$ to $b = 10$, $n = 100$:

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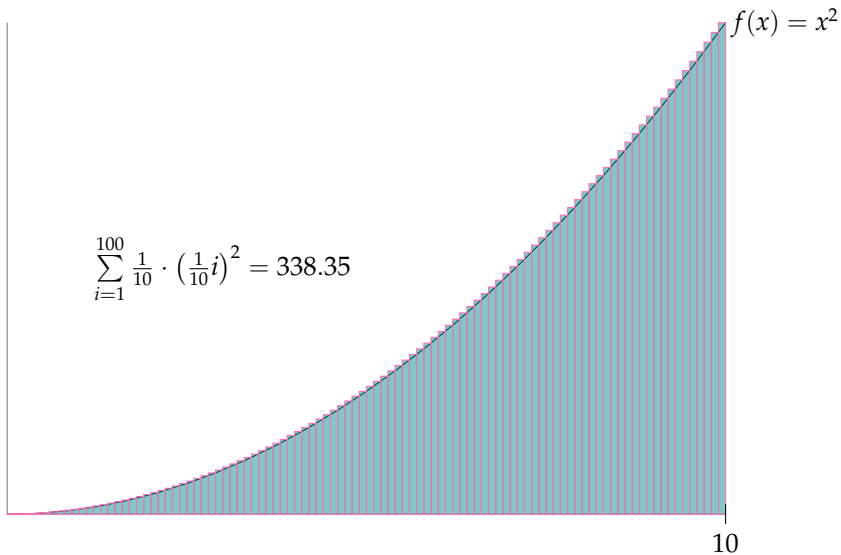
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EVALUATING RIEMANN SUMS IN SIGMA NOTATION

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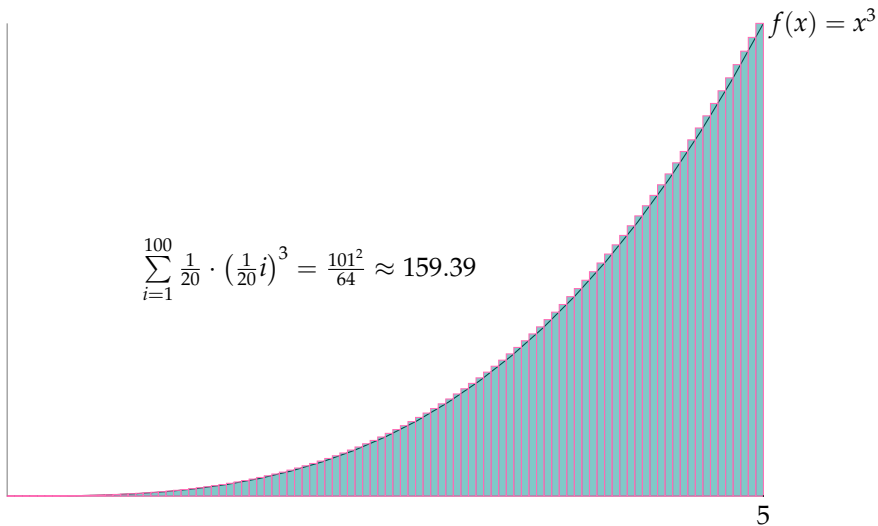
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Definition

Let a and b be two real numbers and let $f(x)$ be a function that is defined for all x between a and b . Then we define $\Delta x = \frac{b-a}{N}$ and

$$\int_a^b f(x) \, dx = \lim_{N \rightarrow \infty} \sum_{i=1}^N f(x_{i,N}^*) \cdot \Delta x$$

when the limit exists and when the choice of $x_{i,N}^*$ in the i^{th} interval doesn't matter.

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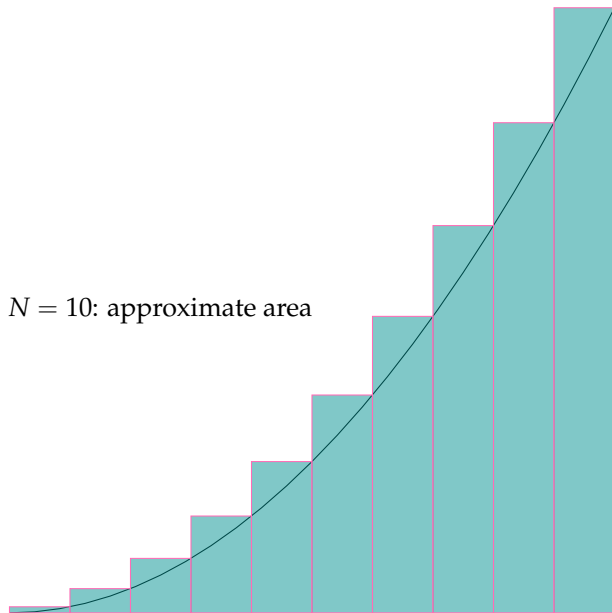
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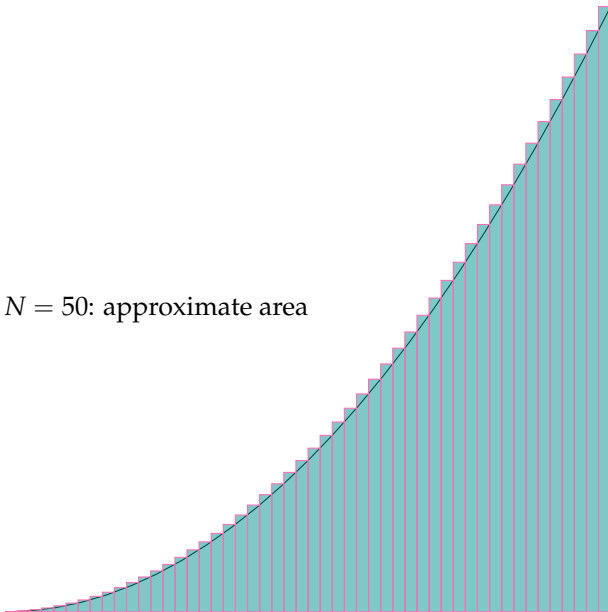
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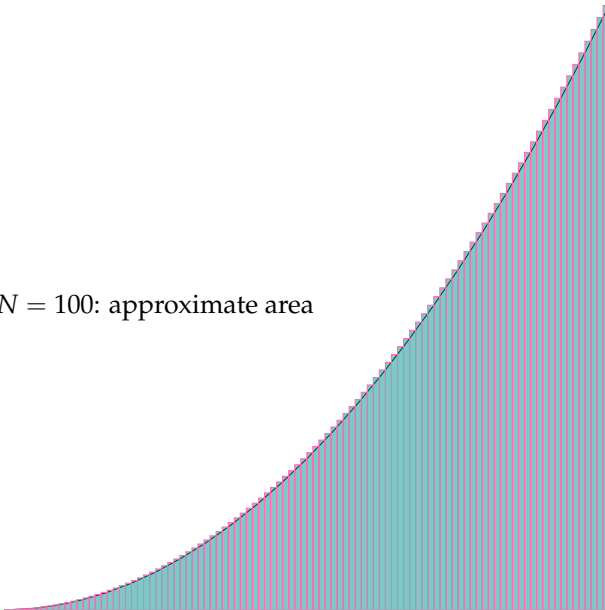
It's understood we're taking a limit as N goes to infinity, so we don't bother specifying N (or each location where we find our height) in the second notation.



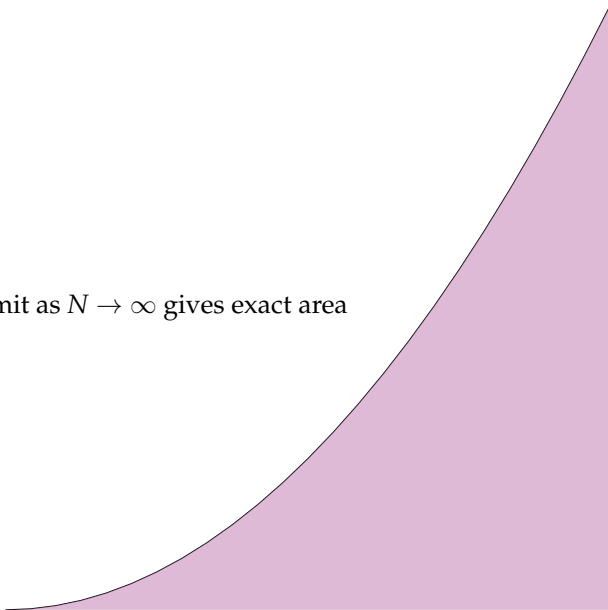
$N = 50$: approximate area



$N = 100$: approximate area



Limit as $N \rightarrow \infty$ gives exact area



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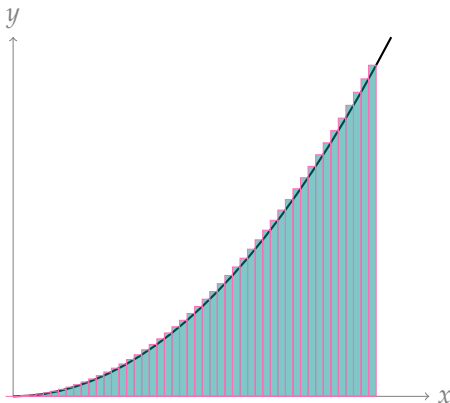
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$$\begin{aligned}\sum_{i=1}^n \Delta x \cdot f(a + i\Delta x) &= \sum_{i=1}^n \frac{5}{n} \cdot \left(\frac{5}{n}i\right)^2 = \sum_{i=1}^n \frac{125}{n^3} i^2 \\ &= \frac{125}{n^3} \left[\sum_{i=1}^n i^2 \right] = \frac{125}{n^3} \left(\frac{n(n+1)(2n+1)}{6} \right) \\ &= \frac{125}{n^2} \left(\frac{(n+1)(2n+1)}{6} \right) = \frac{125}{6} \left(\frac{2n^2 + 3n + 1}{n^2} \right)\end{aligned}$$

We found the right Riemann sum of $y = x^2$ from $a = 0$ to $b = 5$ using n slices:

$$\frac{125}{6} \cdot \frac{2n^2 + 3n + 1}{n^2}$$

Use it to find the exact area under the curve.



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Use it to find the exact area under the curve.

$$\begin{aligned}\int_0^5 x^2 \, dx &= \lim_{n \rightarrow \infty} \left[\frac{125}{6} \cdot \frac{2n^2 + 3n + 1}{n^2} \right] \\ &= \frac{125}{6} \lim_{n \rightarrow \infty} \left[2 + \frac{3}{n} + \frac{1}{n^2} \right] \\ &= \frac{125}{6} (2) = \frac{125}{3}\end{aligned}$$

REFRESHER: LIMITS OF RATIONAL FUNCTIONS

$$\lim_{n \rightarrow \infty} \frac{n^2 + 2n + 15}{3n^2 - 9n + 5} =$$

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REFRESHER: LIMITS OF RATIONAL FUNCTIONS

$$\lim_{n \rightarrow \infty} \frac{n^2 + 2n + 15}{3n^2 - 9n + 5} = \lim_{n \rightarrow \infty} \frac{1 + 2/n + 15/n^2}{3 - 9/n + 5/n^2} = \frac{1}{3}$$

When the degree of the top and bottom are the same, the limit as n goes to infinity is the ratio of the leading coefficients.

$$\lim_{n \rightarrow \infty} \frac{n^2 + 2n + 15}{3n^3 - 9n + 5} =$$

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REFRESHER: LIMITS OF RATIONAL FUNCTIONS

$$\lim_{n \rightarrow \infty} \frac{n^2 + 2n + 15}{3n^2 - 9n + 5} = \lim_{n \rightarrow \infty} \frac{1 + 2/n + 15/n^2}{3 - 9/n + 5/n^2} = \frac{1}{3}$$

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$$\lim_{n \rightarrow \infty} \frac{n^3 + 2n + 15}{3n^2 - 9n + 5} = \lim_{n \rightarrow \infty} \frac{n + 2/n + 15/n^2}{3 - 9/n + 5/n^2} = \infty$$

When the degree of the top is larger than the degree of the bottom, the limit as n goes to infinity is positive or negative infinity.

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}$$

Evaluate $\int_0^1 x^2 \, dx$ exactly using midpoint Riemann sums.

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

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Evaluate $\int_0^1 x^2 dx$ exactly using midpoint Riemann sums.

$$\sum_{i=1}^n \Delta x \cdot \left(\left(i - \frac{1}{2} \right) \Delta x \right)^2$$

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Evaluate $\int_0^1 x^2 dx$ exactly using midpoint Riemann sums.

$$\begin{aligned} \sum_{i=1}^n \Delta x \cdot \left(\left(i - \frac{1}{2} \right) \Delta x \right)^2 &= \frac{1}{n^3} \sum_{i=1}^n \left(i^2 - i + \frac{1}{4} \right) = \frac{1}{n^3} \left[\sum_{i=1}^n i^2 - \sum_{i=1}^n i + \sum_{i=1}^n \frac{1}{4} \right] \\ &= \frac{1}{n^3} \left[\frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} + \frac{1}{4}n \right] \\ &= \frac{2n^2 + 3n + 1}{6n^2} - \frac{n+1}{2n^2} + \frac{1}{4n^2} \end{aligned}$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

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Evaluate $\int_0^1 x^2 dx$ exactly using midpoint Riemann sums.

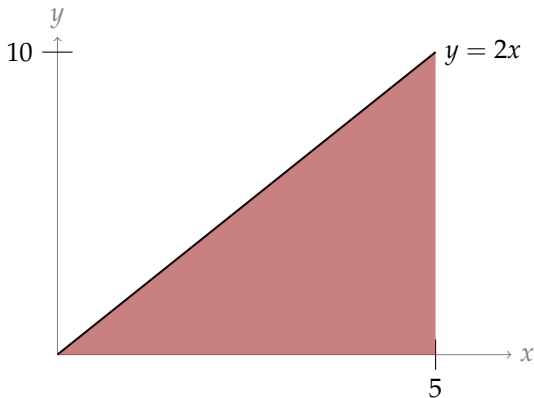
$$\begin{aligned} \sum_{i=1}^n \Delta x \cdot \left(\left(i - \frac{1}{2} \right) \Delta x \right)^2 &= \frac{1}{n^3} \sum_{i=1}^n \left(i^2 - i + \frac{1}{4} \right) = \frac{1}{n^3} \left[\sum_{i=1}^n i^2 - \sum_{i=1}^n i + \sum_{i=1}^n \frac{1}{4} \right] \\ &= \frac{1}{n^3} \left[\frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} + \frac{1}{4}n \right] \\ &= \frac{2n^2 + 3n + 1}{6n^2} - \frac{n+1}{2n^2} + \frac{1}{4n^2} \end{aligned}$$

Exact area under the curve:

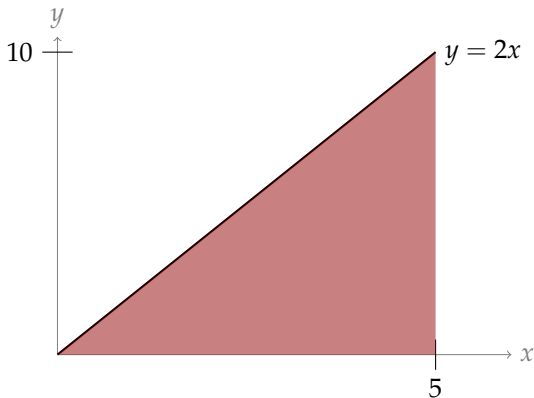
$$\lim_{n \rightarrow \infty} \left[\frac{2n^2 + 3n + 1}{6n^2} - \frac{n+1}{2n^2} + \frac{1}{4n^2} \right] = \frac{2}{6} - 0 + 0 = \frac{1}{3}$$

Let's see some special cases where we can use geometry to evaluate integrals without Riemann sums.

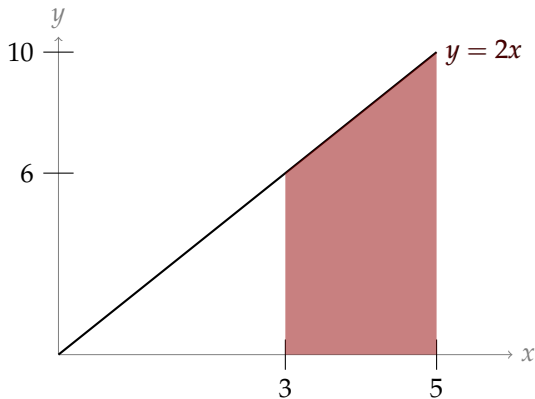
$$\int_0^5 2x \, dx$$



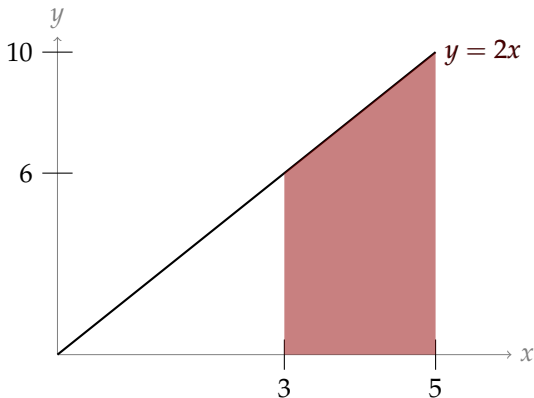
$$\int_0^5 2x \, dx = \frac{1}{2}(5)(10) = 25$$



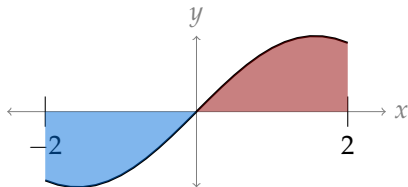
$$\int_3^5 2x \, dx$$



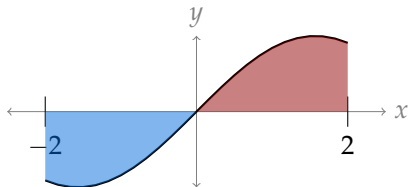
$$\int_3^5 2x \, dx = \frac{1}{2}(5)(10) - \frac{1}{2}(3)(6) = 25 - 9 = 16$$



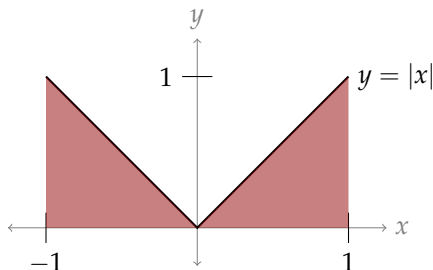
$$\int_{-2}^2 \sin x \, dx$$



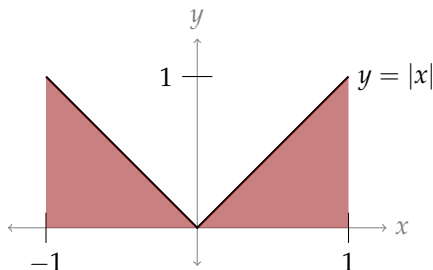
$$\int_{-2}^2 \sin x \, dx = -A + A = 0$$



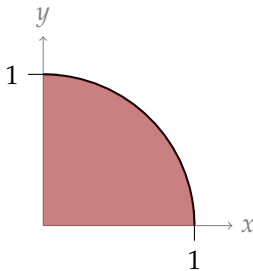
$$\int_{-1}^1 |x| \, dx$$



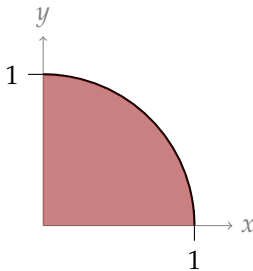
$$\int_{-1}^1 |x| \, dx = \frac{1}{2}(1)(1) + \frac{1}{2}(1)(1) = 1$$



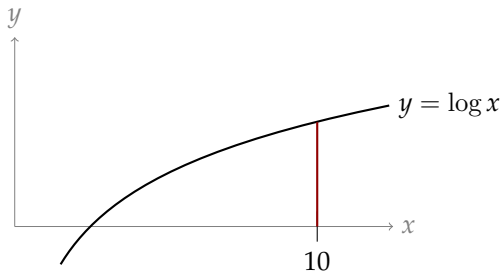
$$\int_0^1 \sqrt{1-x^2} \, dx$$



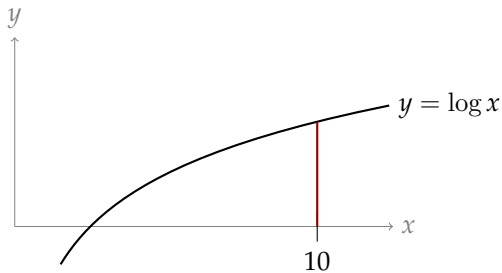
$$\int_0^1 \sqrt{1-x^2} \, dx = \frac{1}{4}(\pi \cdot 1^2) = \frac{\pi}{4}$$



$$\int_{10}^{10} \log x \, dx$$



$$\int_{10}^{10} \log x \, dx = 0$$



A car travelling down a straight highway records the following measurements:

Time	12:00	12:10	12:20	12:30	12:40	12:50	1:00
Speed (kph)	80	100	100	90	90	75	100

Approximately how far did the car travel from 12:00 to 1:00?

A car travelling down a straight highway records the following measurements:

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Approximately how far did the car travel from 12:00 to 1:00?

We don't know the speed of the car over the entire hour, so the best we can do is to use the measured speeds as approximations for the speeds the car travelled over 10-minute intervals.

We can use left, right, and midpoint Riemann sums. Note that there are only six 10-minute intervals, but we know seven points. For a midpoint Riemann sum, since we need to know the speed at the midpoint of the interval, we can only use three intervals, not six.

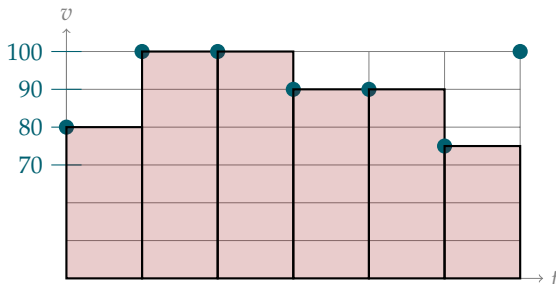
Finally, note that 10 minutes is $\frac{1}{6}$ of an hour, and 20 minutes is $\frac{1}{3}$ of an hour.

A car travelling down a straight highway records the following measurements:

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Approximately how far did the car travel from 12:00 to 1:00?

Left RS: $80 \cdot \underbrace{\frac{1}{6}}_{12:00-12:10} + 100 \cdot \underbrace{\frac{1}{6}}_{12:10-12:20} + 100 \cdot \underbrace{\frac{1}{6}}_{12:20-12:30} + 90 \cdot \underbrace{\frac{1}{6}}_{12:30-12:40} + 90 \cdot \underbrace{\frac{1}{6}}_{12:40-12:50} + 75 \cdot \underbrace{\frac{1}{6}}_{12:50-1:00}$

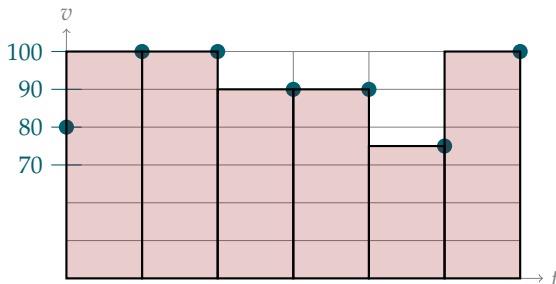


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Approximately how far did the car travel from 12:00 to 1:00?

Right RS: $\underbrace{100 \cdot \frac{1}{6}}_{12:00-12:10} + \underbrace{100 \cdot \frac{1}{6}}_{12:10-12:20} + \underbrace{90 \cdot \frac{1}{6}}_{12:20-12:30} + \underbrace{90 \cdot \frac{1}{6}}_{12:30-12:40} + \underbrace{75 \cdot \frac{1}{6}}_{12:40-12:50} + \underbrace{100 \cdot \frac{1}{6}}_{12:50-1:00}$

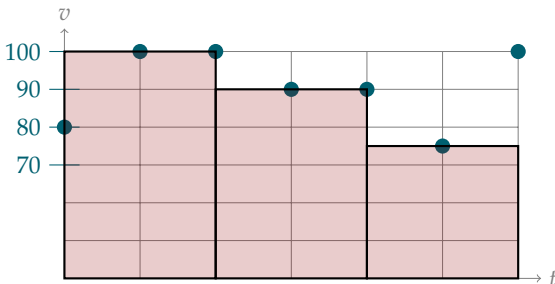


A car travelling down a straight highway records the following measurements:

Time	12:00	12:10	12:20	12:30	12:40	12:50	1:00
Speed (kph)	80	100	100	90	90	75	100

Approximately how far did the car travel from 12:00 to 1:00?

Midpoint RS: $\underbrace{100 \cdot \frac{1}{3}}_{12:00-12:20} + \underbrace{90 \cdot \frac{1}{3}}_{12:20-12:40} + \underbrace{75 \cdot \frac{1}{3}}_{12:40-1:00}$

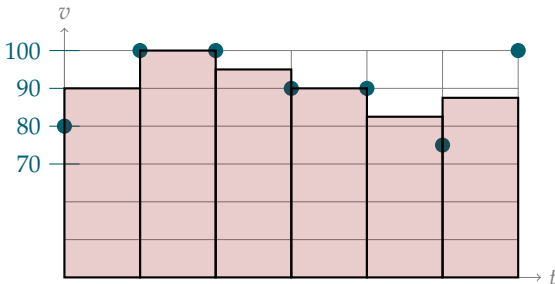


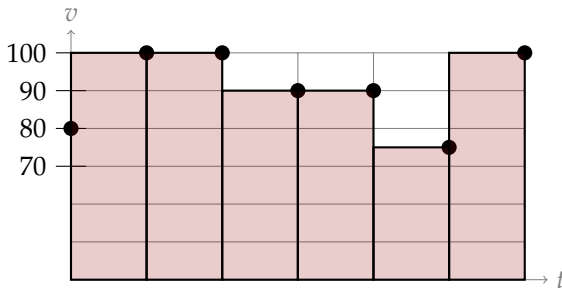
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Time	12:00	12:10	12:20	12:30	12:40	12:50	1:00
Speed (kph)	80	100	100	90	90	75	100

Approximately how far did the car travel from 12:00 to 1:00?

Remark: it's tempting to try to make a midpoint Riemann sum with 6 intervals work by averaging the speeds at the two ends of each interval. This is a perfectly sensible approximation, but it's not a midpoint Riemann sum.





The computation

$$\text{distance} = \text{rate} \times \text{time}$$

looks a lot like the computation

$$\text{area} = \text{base} \times \text{height}$$

for a rectangle. This gives us another interpretation for an integral.

ANOTHER INTERPRETATION OF THE INTEGRAL

Let $x(t)$ be the position of an object moving along the x -axis at time t , and let $v(t) = x'(t)$ be its velocity. Then for all $b > a$,

$$x(b) - x(a) = \int_a^b v(t) \, dt$$

That is, $\int_a^b v(t) \, dt$ gives the *net distance* moved by the object from time a to time b .

Included Work

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